

AMBA and AXI Connect [Servo Motor] Example

Micro Servo Module

[Reference] ■

➤ SoC Peripheral RTC Design Project (프로젝트 최종 계획서 제출)

1. Project Exam

1) Exam_Washing_Machine

2) Exam_Microblaze_Washing_Machine_Block_Design

2. led_btn_Exam

3. AXI_My_IP_LED_Exam

4. AXI_My_IP_FND_Exam

5. AXI_RTC_Exam

6. AXI_PWM

7. AXI_LCD

8. Servo_Motor_Exam

9. DHT11_Module_Exam

10. Ultrasonic Module Exam

AMBA and AXI Connect [Servo Motor] Example

Micro Servo Module

[Create IP]

➤ AXI Connect ➔ AXI Template



New IP Location



➤ *AXI Connect* ➔ *AXI Template*



Create a New Customized IP Location

This wizard will guide you through the creation of a new customized IP location

The Manage IP flow requires a directory for your IP files to be specified. Additional options you will be able to specify include the targeted part, preferred language for IP files as well as supported simulator.



< Back

Next >

Finish

Cancel

➤ AXI Connect ➔ AXI Template

New IP Location

Manage IP Settings
Set options for creating and generating IP.

Part 수정

Part: xc7vx485tffg1157-1 (active) ...

Target language: Verilog

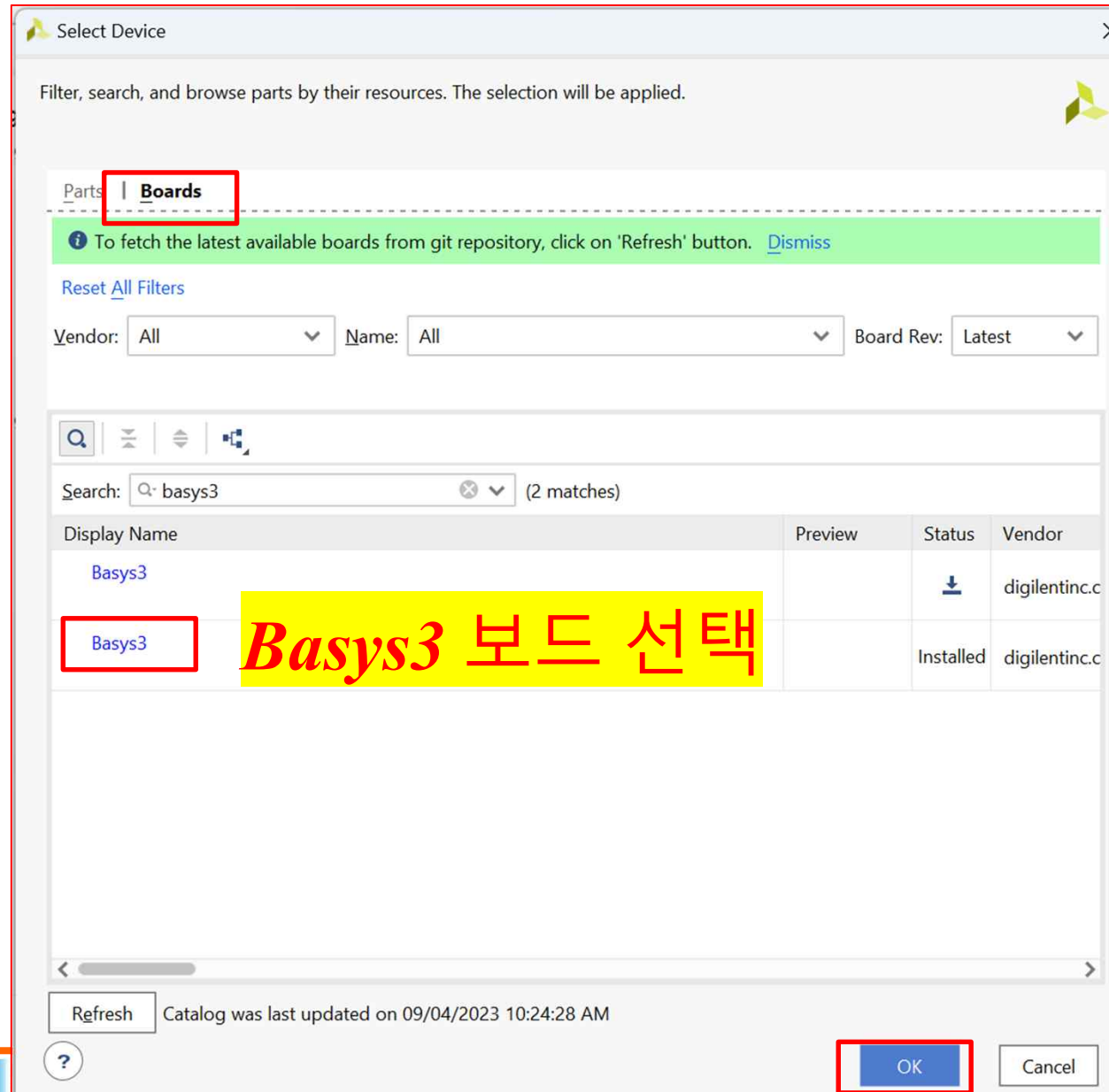
Target simulator: Vivado Simulator

Simulator language: Mixed

IP location: C:/Users/parkj ...

< Back Next > Finish Cancel

➤ AXI Connect ➔ AXI Template



New IP Location

➤ **AXI Connect ➔ AXI Template**

Manage IP Settings

Set options for creating and generating IP.

Part: Basys3 (xc7a35tcpg236-1)

Target language: Verilog

Target simulator: Vivado Simulator

Simulator language: Mixed

IP location: C:/Users/parkj

원하는 *IP location* 지정

Finish



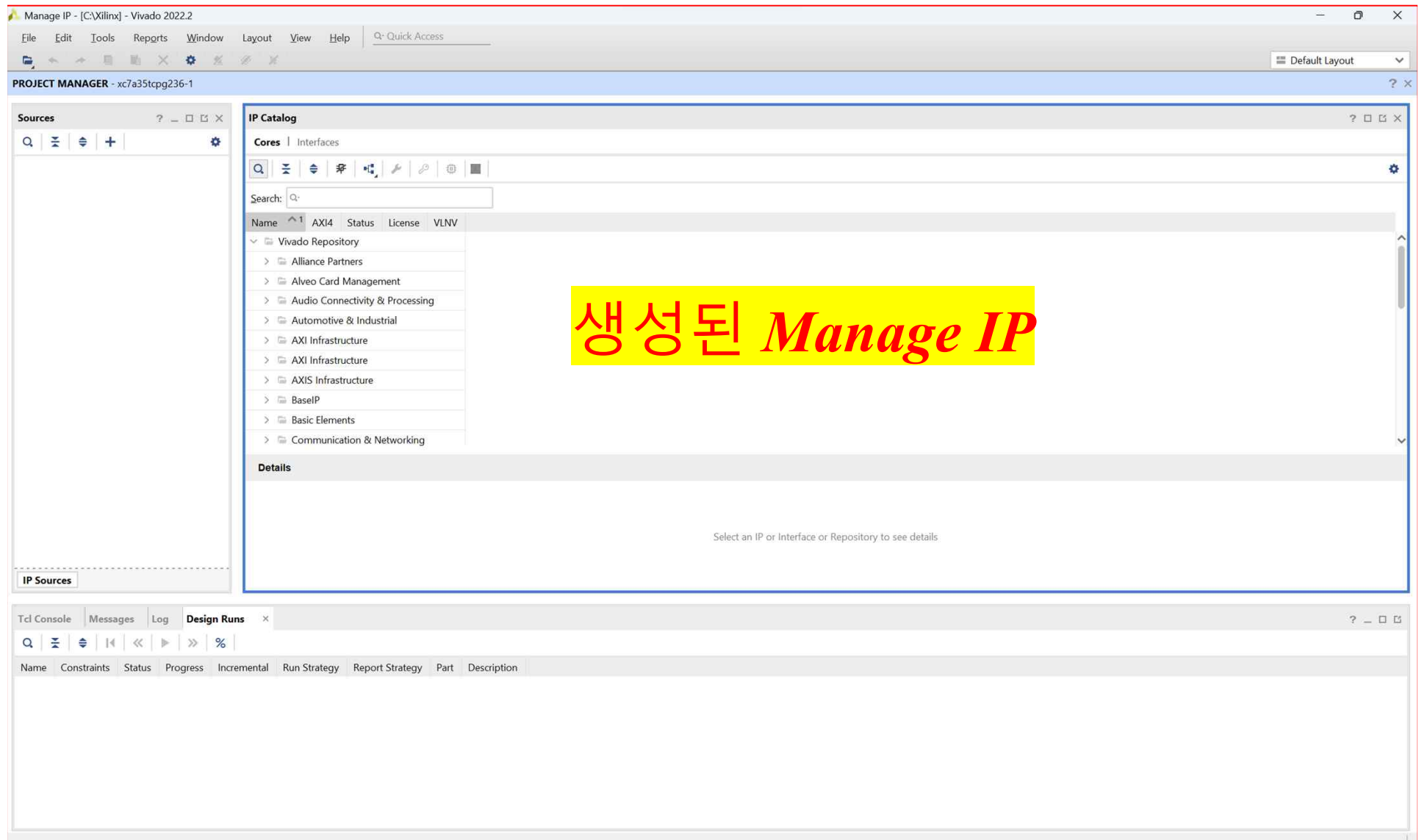
< Back

Next >

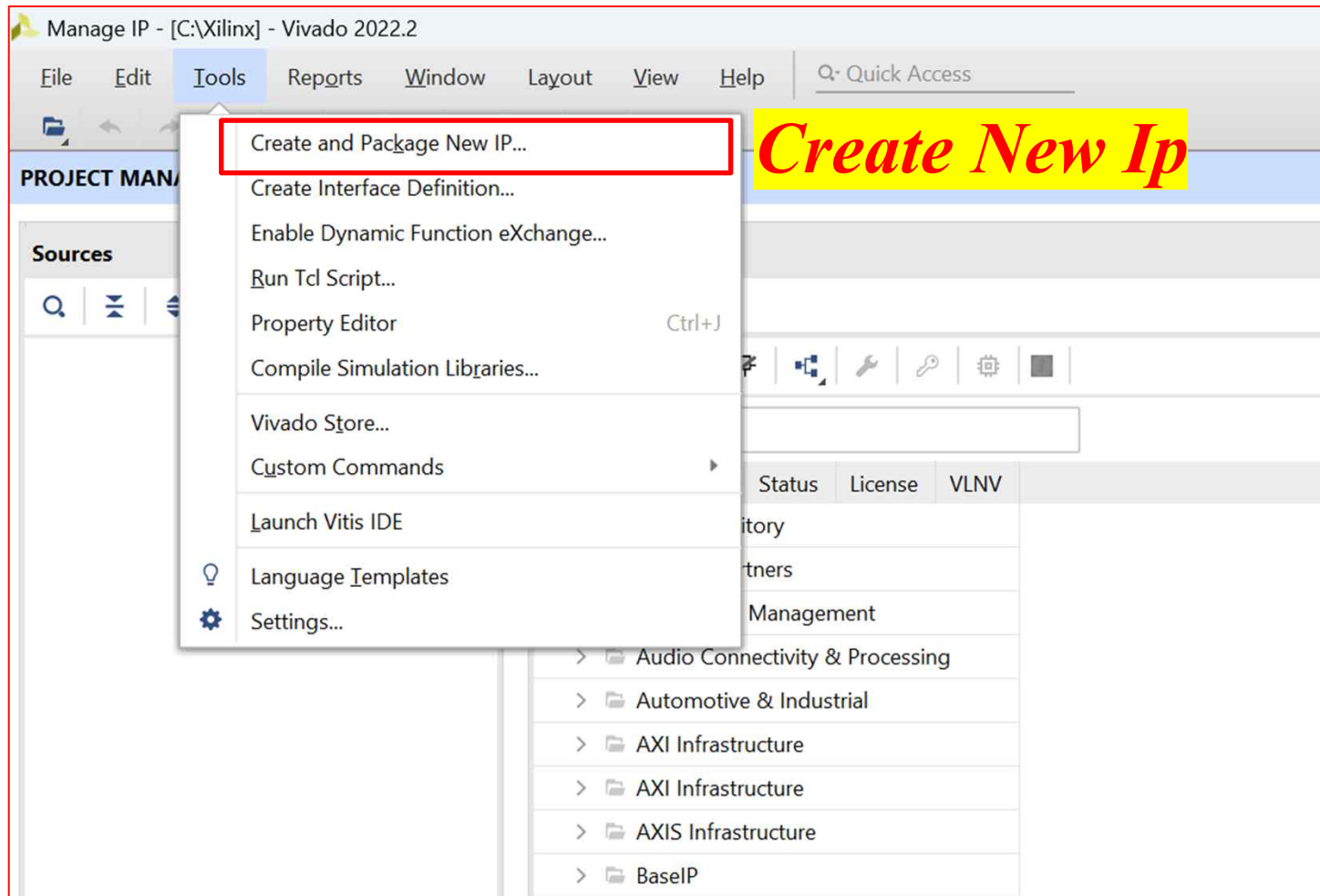
Finish

Cancel

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Create and Package New IP

Create Peripheral, Package IP or Package a Block Design
Please select one of the following tasks.

Packaging Options

- ☐ Package your current project
Use the project as the source for creating a new IP Definition.
- ☐ Package a block design from the current project
Choose a block design as the source for creating a new IP Definition.
Select a block design:
- ☐ Package a specified directory
Choose a directory as the source for creating a new IP Definition.

Create AXI4 Peripheral

- ☒ Create a new AXI4 peripheral
Create an AXI4 IP, driver, software test application, IP Integrator AXI4 VIP simulation and debug demonstration design.

Create New Ip

➤ AXI Connect ➔ AXI Template

Create and Package New IP

Peripheral Details
Specify name, version and description for the new peripheral

Name: servo_axi

Version: 1.0

Display name: servo_axi_v1.0

Description: My new AXI IP

IP location: C:/Xilinx/managed_ip_project/./ip_repo

☐ Overwrite existing

이름 변경 후 *Next*

< Back **Next >** Finish Cancel

➤ AXI Connect ➔ AXI Template

Create and Package New IP

Add Interfaces
Add AXI4 interfaces supported by your peripheral

☐ Enable Interrupt Support

Interfaces
S00_servo

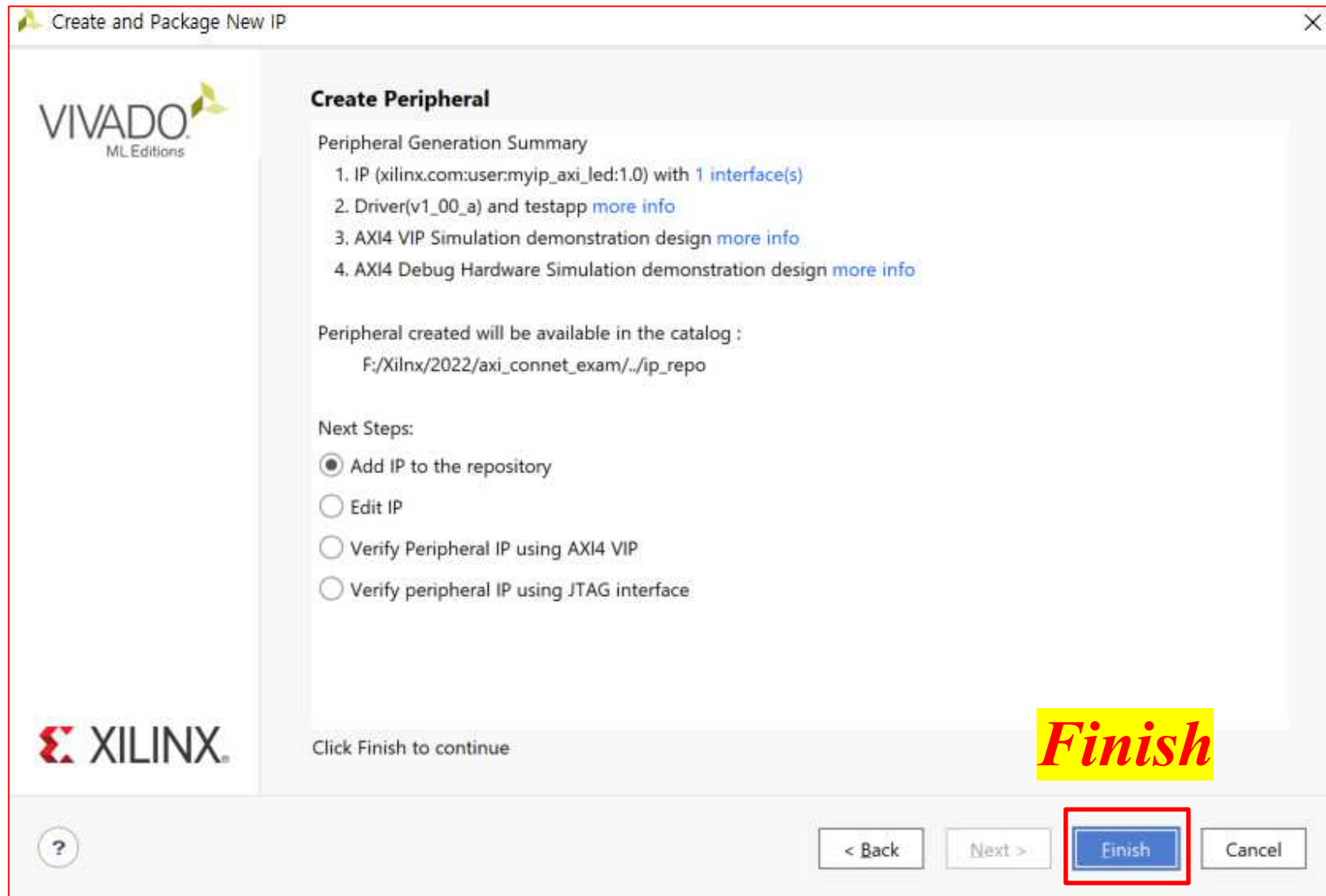
S00_servo_AXI
servo_axi_v1.0

Name	S00_servo_AXI
Interface Type	Lite
Interface Mode	Slave
Data Width (Bits)	32
Memory Size (Bytes)	64
Number of Registers	4 [4..512]

이름 변경 후 *Next*

< Back Next > Finish Cancel

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

IP Catalog

Cores | Interfaces

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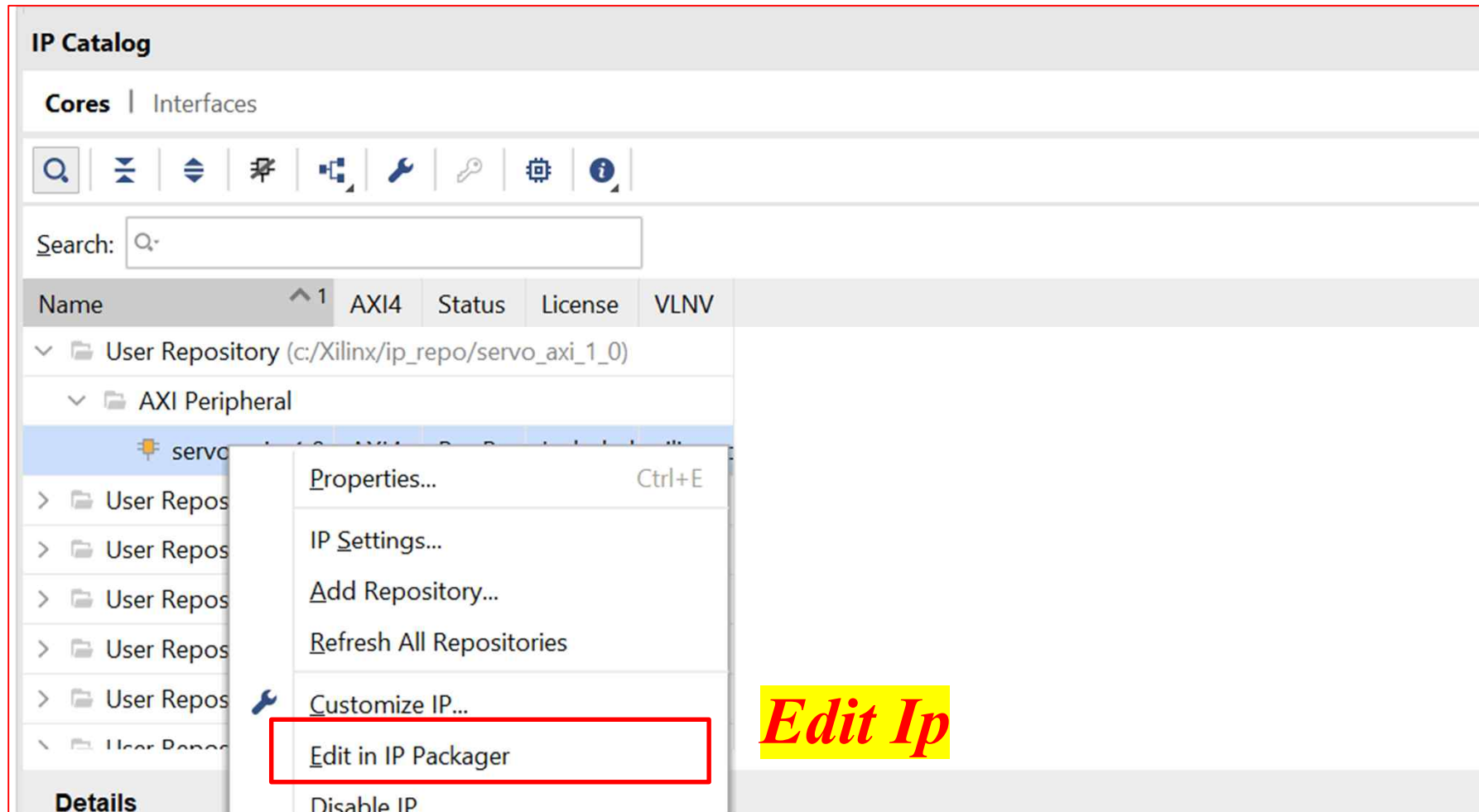
≡

Search:

Name	AXI4	Status	License	VLNV
User Repository (c:/Xilinx/ip_repo/servo_axi_1_0) AXI Peripheral servo_axi_v1.0 AXI4 Pre-Proc Included xilinx.cc				
> User Repository (c:/Xilinx/ip_repo/dht11_1_0)				
> User Repository (c:/Xilinx/ip_repo/ultrasonic_axi_1_0)				
> User Repository (c:/Xilinx/ip_repo/pwm_axi_1_0)				
> User Repository (c:/Xilinx/ip_repo/lcd_axi_1_0)				
> User Repository (c:/Xilinx/ip_repo/lcd1602_axi_1_0)				
> User Repository (c:/Xilinx/ip_repo/pwm_axi_base_1_0)				

IP 생성 확인
(ip_repo에 저장)

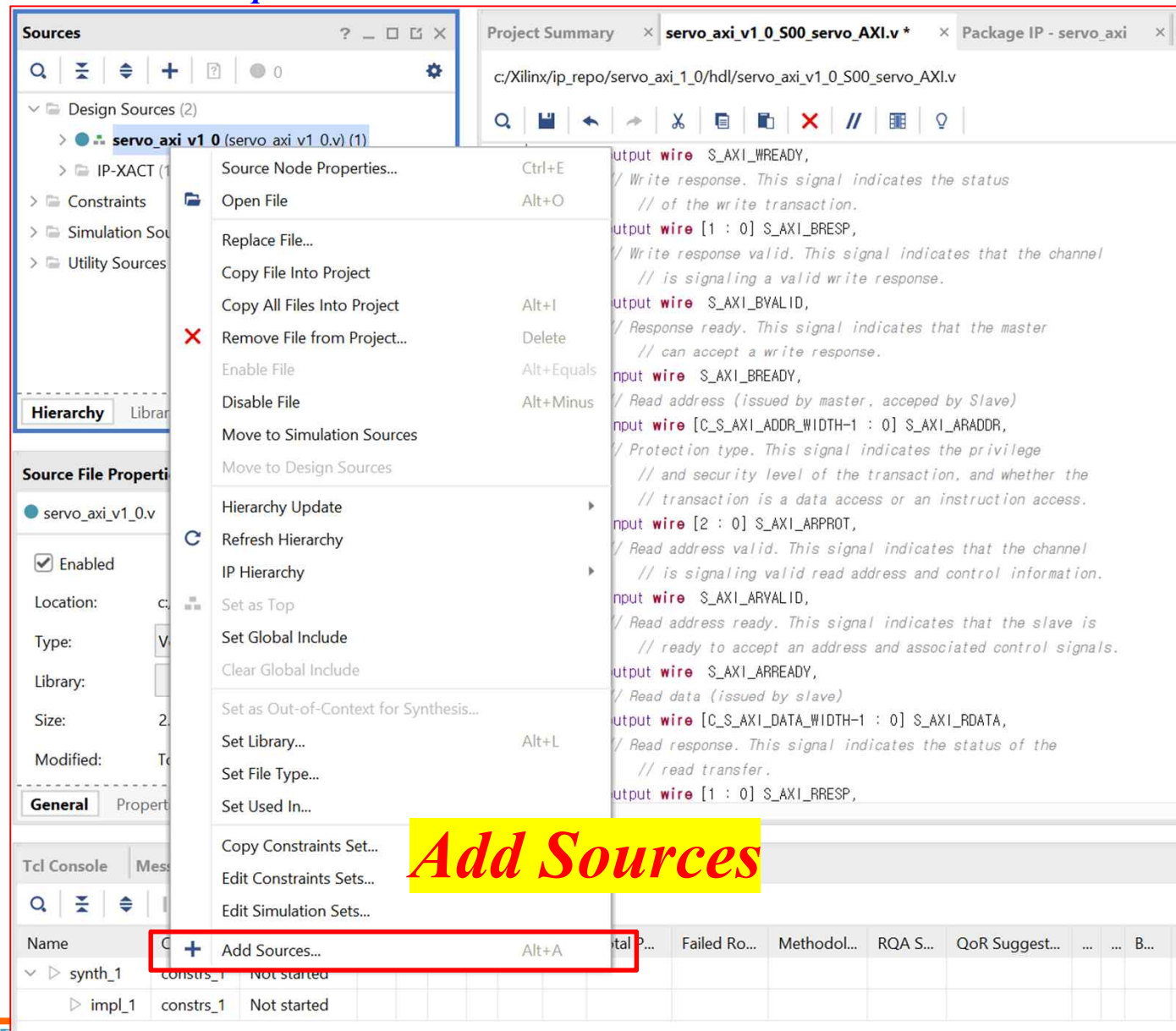
➤ AXI Connect ➔ AXI Template



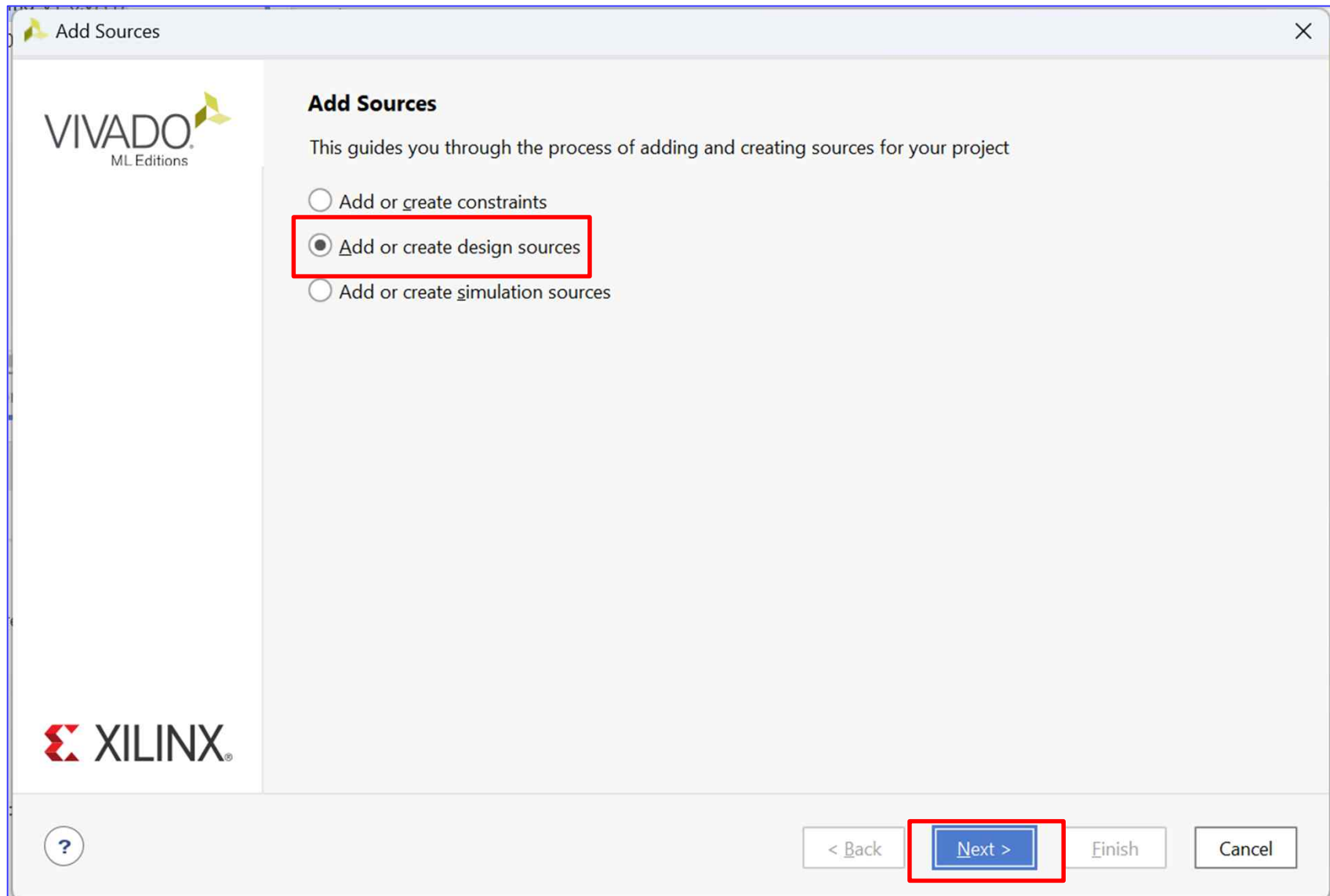
➤ AXI Connect ➔ AXI Template

The screenshot displays the Xilinx IDE interface. On the left, the 'Sources' window shows a project hierarchy with 'Design Sources (2)' containing 'servo_axi_v1_0 (servo_axi_v1_0.v) (1)' and 'IP-XACT (1)'. A red box highlights the 'servo_axi_v1_0' entry. Below it, a yellow box contains the Korean text '저장된 IP 모듈 확인' (Check saved IP module). The 'Properties' window is empty, showing 'Select an object to see properties'. The 'Project Summary' window is open, showing a list of 'Packaging Steps' with green checkmarks: Identification, Compatibility, File Groups, Customization Parameters, Ports and Interfaces, Addressing and Memory, Customization GUI, and Review and Package. The 'Package IP - servo_axi' window is also open, showing the 'Identification' tab with fields for Vendor (xilinx.com), Library (user), Name (servo_axi), Version (1.0), Display name (servo_axi_v1.0), Description (My new AXI IP), Vendor display name, Company url, Root directory (c:/Xilinx/ip_repo/servo_axi_1_0), and Xml file name (c:/Xilinx/ip_repo/servo_axi_1_0/comp). The 'Categories' section shows a list with 'AXI_Peripheral'.

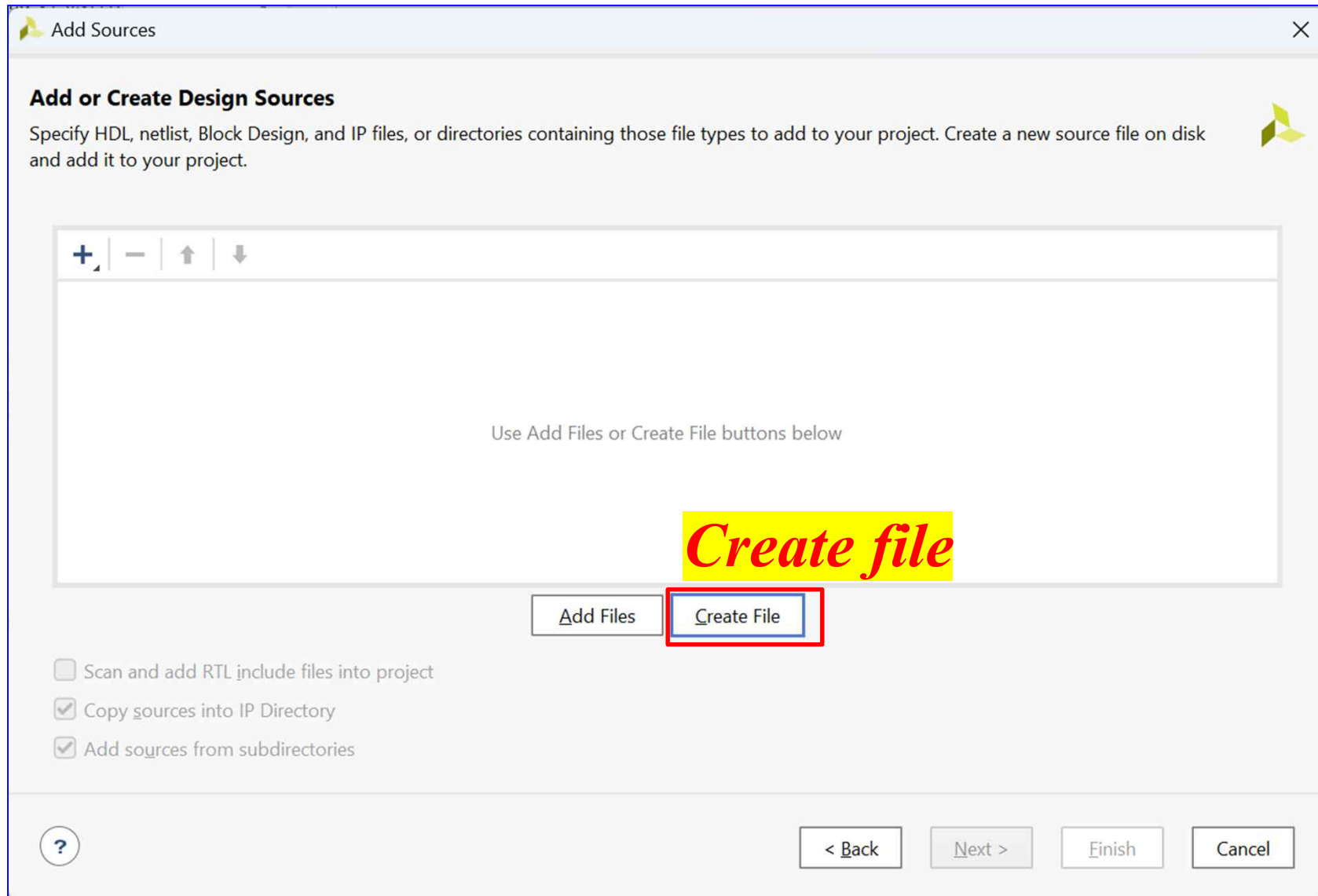
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Add Sources

Add or Create Design Sources

Specify HDL, netlist, Block Design, and IP files, or directories containing those file types to add to your project. Create a new source file on disk and add it to your project.

	Index	Name	Library	Location
●	1	servo.v	xil_defaultlib	C:/Users/parkj/231221_servo
●	2	clockdivider_10000hz.v	xil_defaultlib	C:/Users/parkj/231221_servo

Source 추가

☐ Scan and add RTL include files into project

☒ Copy sources into IP Directory

☒ Add sources from subdirectories

```

1. `timescale 1ns / 1ps
2. module servo(
3.   input clk, reset,
4.   input clk10000hz,
5.   input [7:0] data,
6.   output reg Servo
7. );

8.   reg [1:0] mode;
9.   reg [7:0] duty;
10.  reg [9:0] cnt_duty;

11. always @(posedge clk or negedge reset) begin
12.   if(~reset) begin
13.    duty = 0;
14.   end else begin
15.    duty = data;
16.   end
17. end

18. always @(posedge clk10000hz or negedge reset) begin
19.   if(~reset) begin
20.    cnt_duty = 0;
21.    Servo = 0;
22.   end else begin
23.
24.    if(cnt_duty >= 200) cnt_duty = 0;
25.    else cnt_duty = cnt_duty + 1;
26.
27.    if(cnt_duty < duty) Servo = 1;
28.    else Servo = 0;
29.   end
30. end

31. endmodule

```

```

1. `timescale 1ns / 1ps
2. module clockdivider_10000hz (
3.   input clk,
4.   input reset,
5.   output reg clk10000hz
6. );
7.   reg [25:0] cnt = 0;
8.   always @(posedge clk, negedge reset) begin
9.     if (~reset) begin
10.      cnt <= 0;
11.     end else begin
12.      if (cnt == (5_000 - 1)) begin
13.       cnt <= 0;
14.       clk10000hz <= ~clk10000hz;
15.      end else begin
16.       cnt <= cnt + 1;
17.      end
18.     end
19.   end
20. endmodule

```

➤ AXI Connect ➔ AXI Template

```

Project Summary x Package IP - servo_axi x servo_axi_v1_0.v * x s
c:/Xilinx/ip_repo/servo_axi_1_0/hdl/servo_axi_v1_0.v

16 (
17 // Users to add ports here
18 output wire Servo,
19 // User ports ends
20 // Do not modify the ports beyond this line
21
22
23 // Ports of Axi Slave Bus Interface S00 servo AXI
    
```

```

Project Summ 추가 ro_axi_v1_0_S00_servo_AXI.v * x Package
c:/Xilinx/ip_repo/servo_axi_1_0/hdl/servo_axi_v1_0_S00_servo_AXI.v

13 // Width of S_AXI address bus
14 parameter integer C_S_AXI_ADDR_WIDTH = 4
15 )
16 (
17 // Users to add ports here
18 output wire Servo,
    
```

```

Project Summary x servo_axi_v1_0.v * x servo_axi_v
c:/Xilinx/ip_repo/servo_axi_1_0/hdl/servo_axi_v1_0.v

55 .S_AXI_AWVALID(s00_servo_axi_awvalid),
56 .S_AXI_AWREADY(s00_servo_axi_awready),
57 .S_AXI_WDATA(s00_servo_axi_wdata),
58 .S_AXI_WSTRB(s00_servo_axi_wstrb),
59 .S_AXI_WVALID(s00_servo_axi_wvalid),
60 .S_AXI_WREADY(s00_servo_axi_wready),
61 .S_AXI_BRESP(s00_servo_axi_bresp),
62 .S_AXI_BVALID(s00_servo_axi_bvalid),
63 .S_AXI_BREADY(s00_servo_axi_bready),
64 .S_AXI_ARADDR(s00_servo_axi_araddr),
65 .S_AXI_ARPROT(s00_servo_axi_arprot),
66 .S_AXI_ARVALID(s00_servo_axi_arvalid),
67 .S_AXI_ARREADY(s00_servo_axi_arready),
68 .S_AXI_RDATA(s00_servo_axi_rdata),
69 .S_AXI_RRESP(s00_servo_axi_rresp),
70 .S_AXI_RVALID(s00_servo_axi_rvalid),
71 .S_AXI_RREADY(s00_servo_axi_rready),
72 .Servo(Servo)
73 );
74
75 // Add user logic here
76
77 // User logic ends
78
    
```

AMBA and AXI Connect [Servo Motor]

➤ AXI Connect ➔ AXI Template

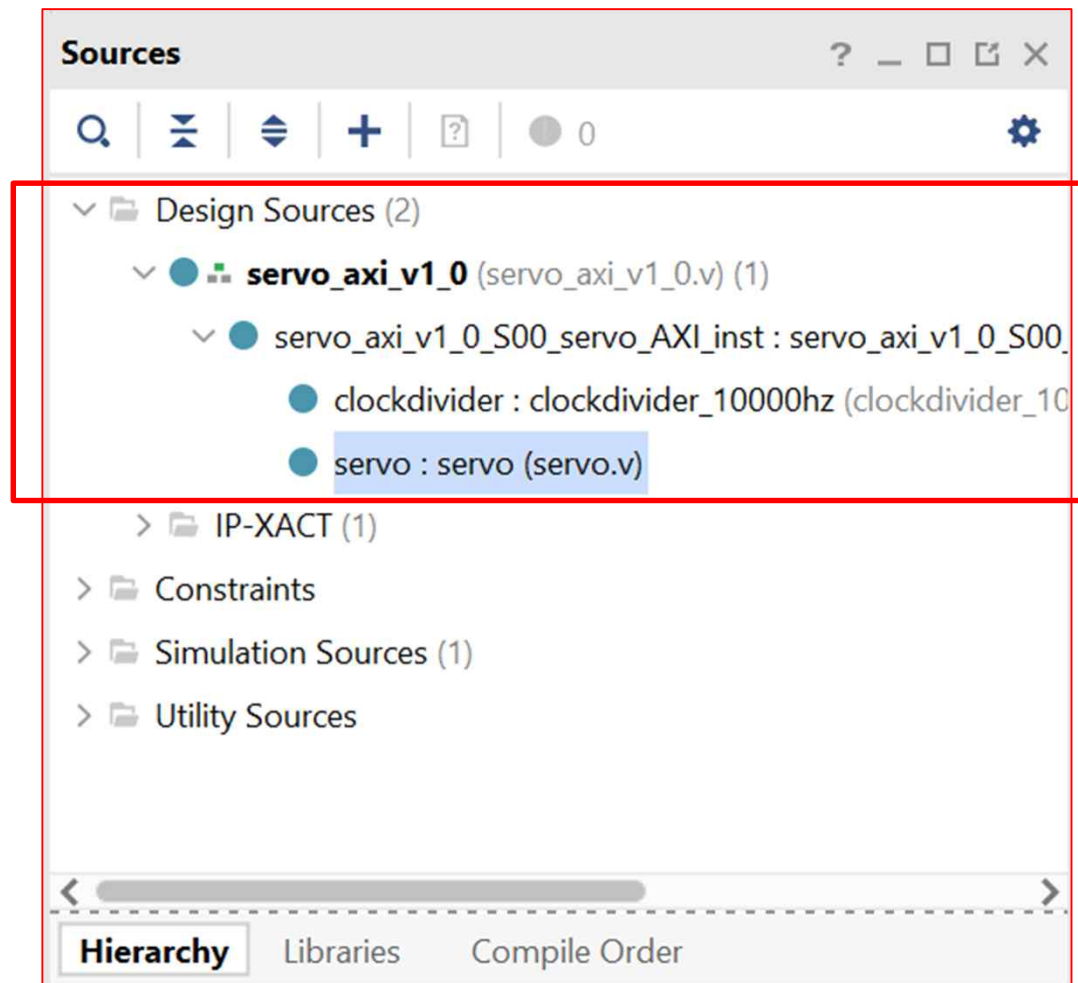
Project Summary x servo_axi_v1_0_S00_servo_AXI.v x Exam.xpr

c:/Xilinx/ip_repo/servo_axi_1_0/hdl/servo_axi_v1_0_S00_servo_AXI.v

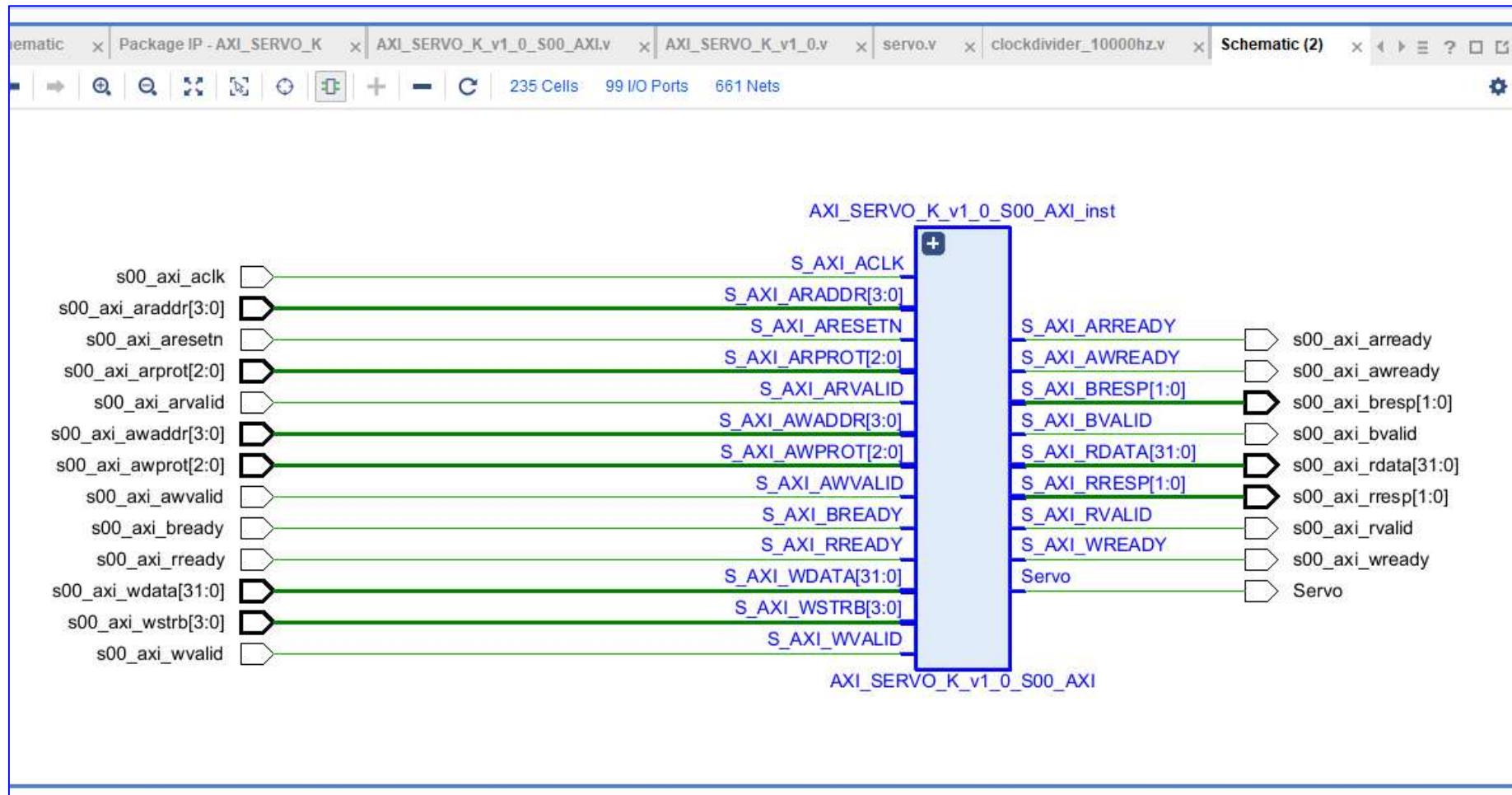
```
391      // acceptance of read address by the slave
392      // output the read data
393      if (slv_reg_rden)
394      begin
395          axi_rdata <= reg_data_out;    // regis
396      end
397  end
398  end
399
400      // Add user logic here
401
402  clockdivider_10000hz clockdivider(
403      .clk(S_AXI_ACLK),
404      .reset(S_AXI_ARESETN),
405      .clk10000hz(w_clkout)
406  );
407
408  servo servo(
409      .clk(S_AXI_ACLK),
410      .reset(S_AXI_ARESETN),
411      .clk10000hz(w_clkout),
412      .Servo(Servo),
413      .data(data)
414  );
415
416  wire [7:0] data;
417  assign data = slv_reg0[7:0];
418  // User logic ends
```

추가

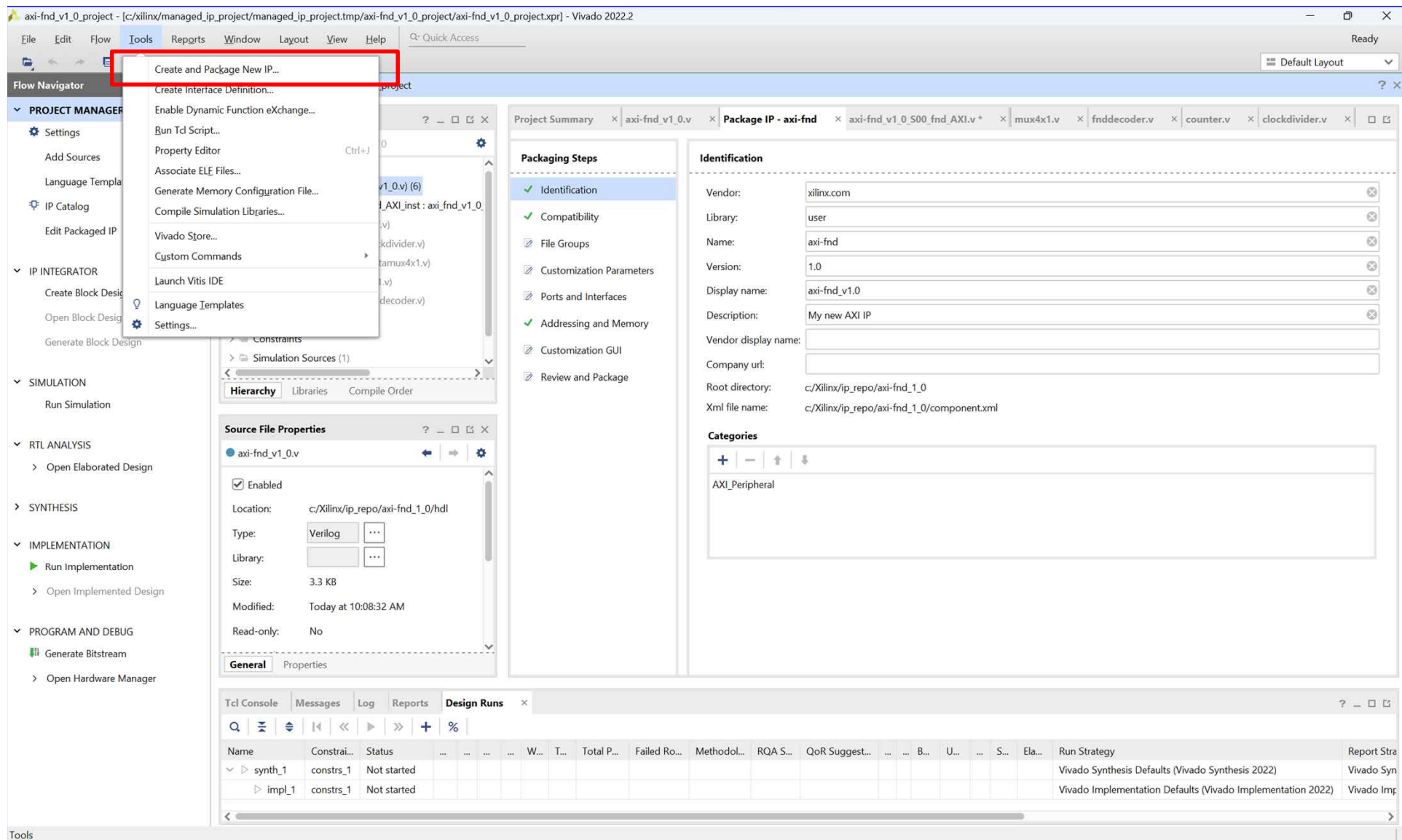
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



The screenshot shows the Vivado 2022.2 IDE interface. The 'Tools' menu is open, and the 'Create and Package New IP...' option is highlighted with a red rectangle. The 'Package IP - axi-fnd' dialog box is visible, showing the 'Identification' tab. The 'Identification' tab contains the following fields:

- Vendor: xilinx.com
- Library: user
- Name: axi-fnd
- Version: 1.0
- Display name: axi-fnd_v1.0
- Description: My new AXI IP
- Vendor display name:
- Company url:
- Root directory: c:/Xilinx/ip_repo/axi-fnd_1_0
- Xml file name: c:/Xilinx/ip_repo/axi-fnd_1_0/component.xml

The 'Categories' section shows 'AXI_Peripheral'.

The 'Packaging Steps' section shows the following steps:

- Identification (checked)
- Compatibility (checked)
- File Groups (checked)
- Customization Parameters (checked)
- Ports and Interfaces (checked)
- Addressing and Memory (checked)
- Customization GUI (checked)
- Review and Package (checked)

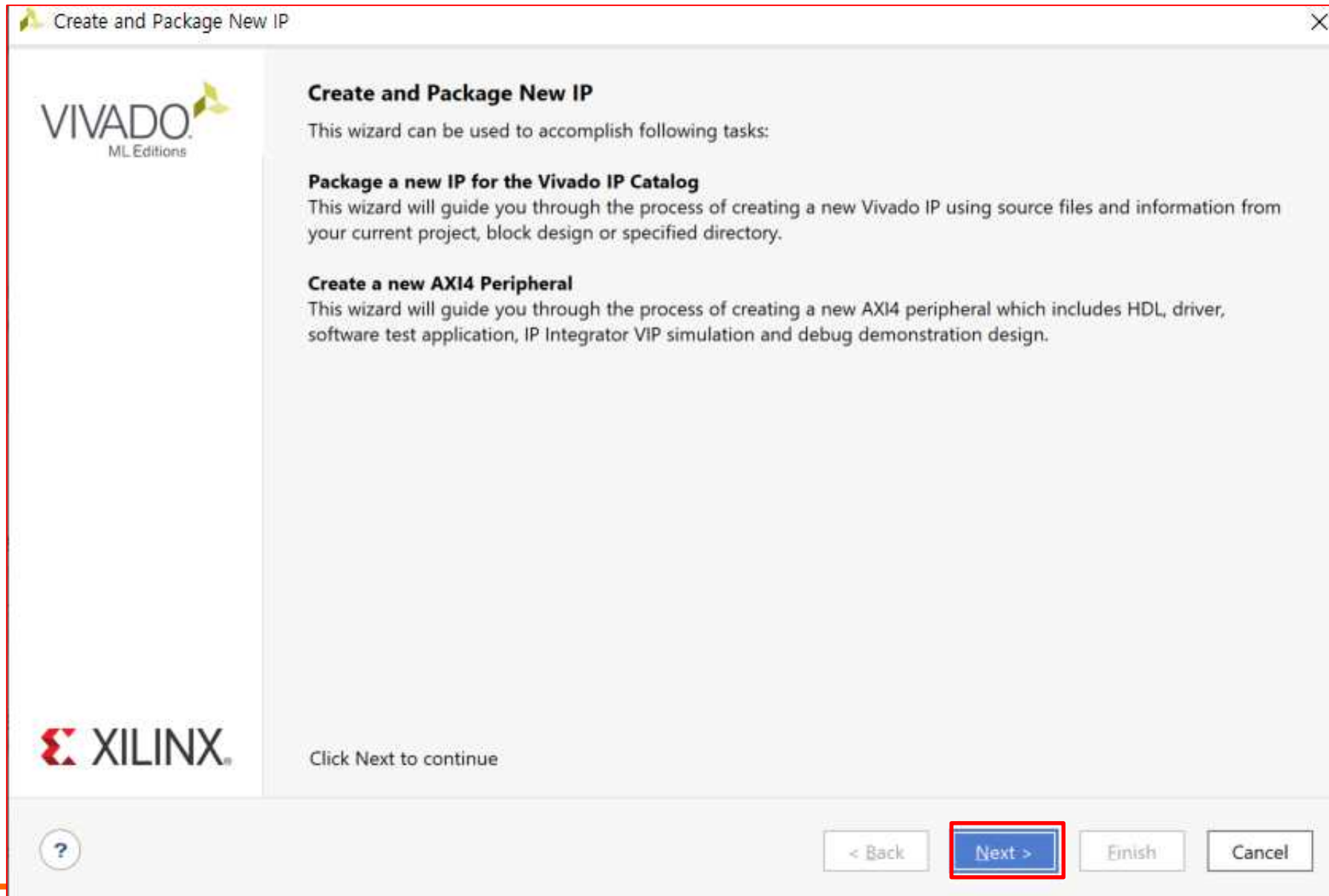
The 'Source File Properties' window shows the following details for 'axi-fnd_v1_0.v':

- Location: c:/Xilinx/ip_repo/axi-fnd_1_0/hdl
- Type: Verilog
- Library:
- Size: 3.3 KB
- Modified: Today at 10:08:32 AM
- Read-only: No

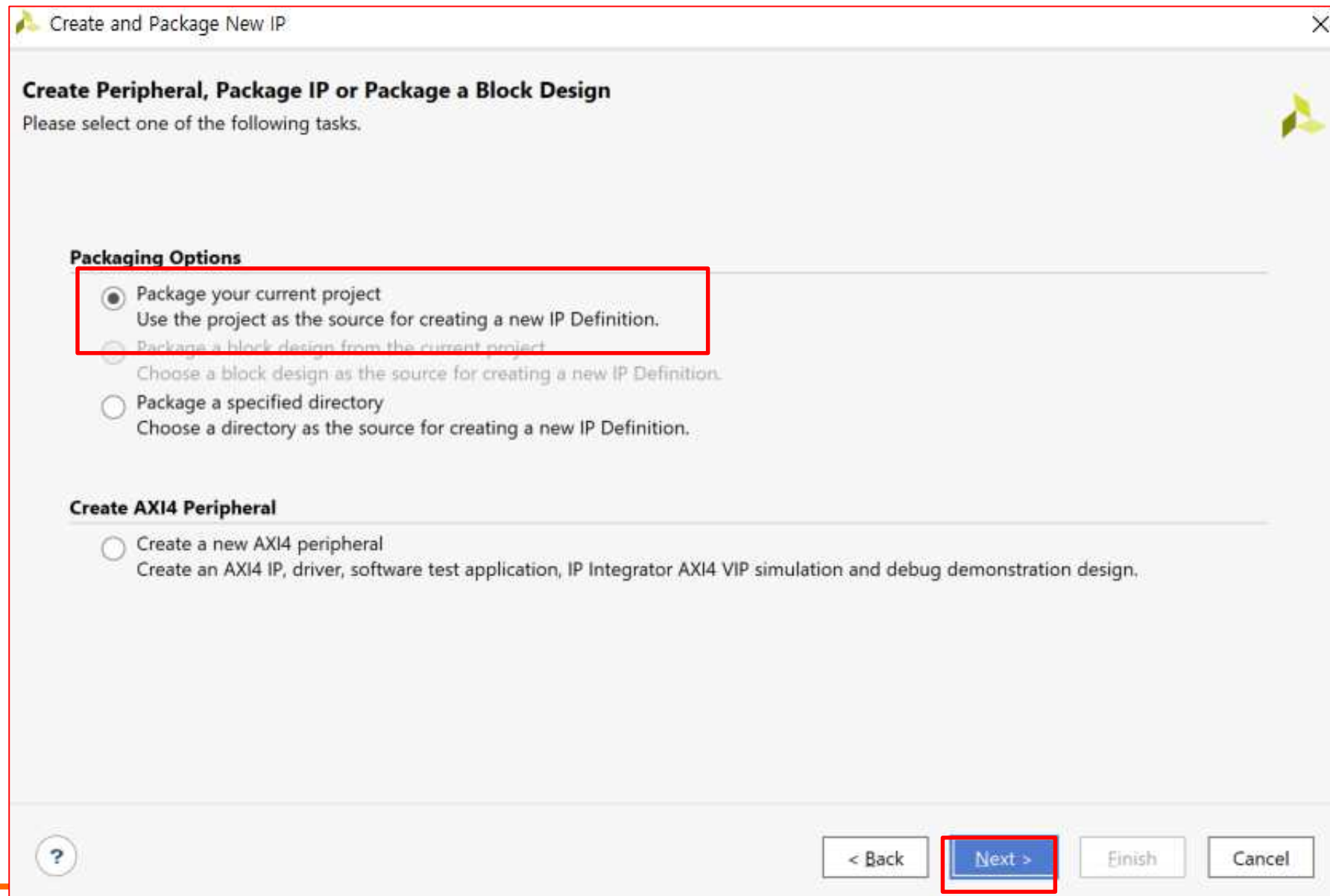
The 'Design Runs' table at the bottom shows the following data:

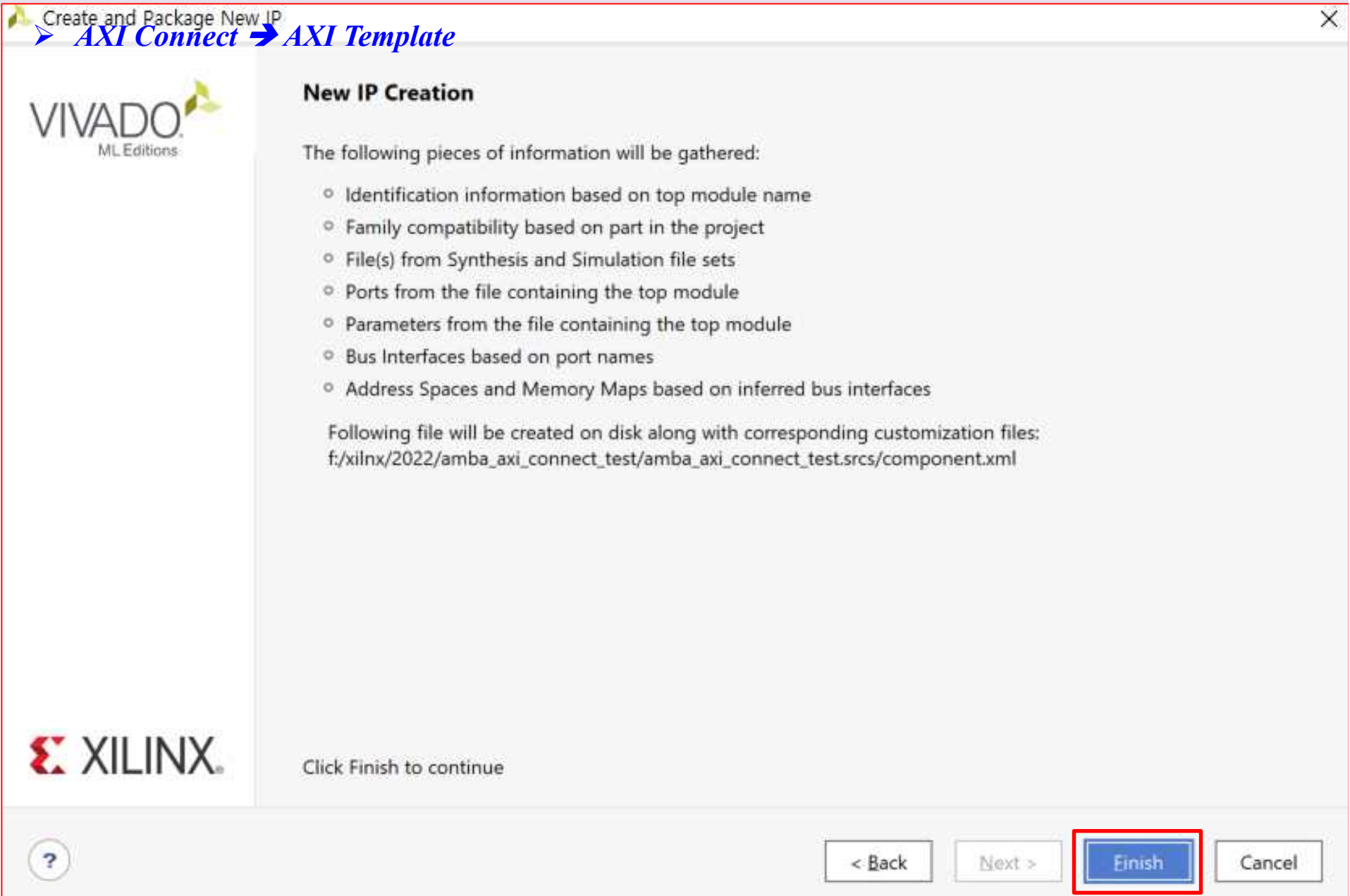
Name	Constrai...	Status	...	...	...	W...	T...	Total P...	Failed Ro...	Methodol...	RQA S...	QoR Suggest...	...	...	B...	U...	S...	Ela...	Run Strategy	Report Stra
synth_1	constrs_1	Not started																	Vivado Synthesis Defaults (Vivado Synthesis 2022)	Vivado Syn
impl_1	constrs_1	Not started																	Vivado Implementation Defaults (Vivado Implementation 2022)	Vivado Imp

➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template





➤ AXI Connect ➔ AXI Template

MyIP 생성

The screenshot displays the Xilinx Project Manager interface for a project named 'fnd_axi_v1_0_project'. The 'Sources' pane on the left shows the project hierarchy, including 'Design Sources (2)', 'IP-XACT (1)', 'Constraints', 'Simulation Sources (1)', and 'Utility Sources'. The 'Properties' pane at the bottom left is empty, showing 'Select an object to see properties'.

The main window is titled 'Package IP - fnd_axi' and contains the 'Packaging Steps' pane on the left, which lists the following steps with green checkmarks:

- Identification
- Compatibility
- File Groups
- Customization Parameters
- Ports and Interfaces
- Addressing and Memory
- Customization GUI
- Review and Package

The 'Identification' tab is active, showing the following fields:

- Vendor: xilinx.com
- Library: user
- Name: fnd_axi
- Version: 1.0
- Display name: fnd_axi_v1.0
- Description: My new AXI IP
- Vendor display name: (empty)
- Company url: (empty)
- Root directory: c:/Xilinx/ip_repo/fnd_axi_1_0
- Xml file name: c:/Xilinx/ip_repo/fnd_axi_1_0/component.xml

The 'Categories' section at the bottom shows a list with the entry 'AXI_Peripheral'.

AMBA and AXI Connect Example

Servo Motor Module

[*Create Project* ➔ *Create Block Design*]

➤ AXI Connect ➔ AXI Template

Settings 클릭

Project Summary

Overview | Dashboard

Settings [Edit](#)

Project name: project_1
 Project location: C:/Users/parkj/test231013/project_1
 Product family: Artix-7
 Project part: Basys3 (xc7a35tcbg236-1)
 Top module name: Not defined
 Target language: Verilog
 Simulator language: Mixed

Board Part

Display name: Basys3
 Board part name: digilentinc.com:basys3:part0:1.2
 Board revision: C.0
 Connectors: No connections
 Repository path: E:/Xilinx/Vivado/2022.2/data/boards/board_files
 URL: <https://digilent.com/reference/programmable-logic/basys-3/start>
 Board overview: Basys3

Synthesis **Implementation**

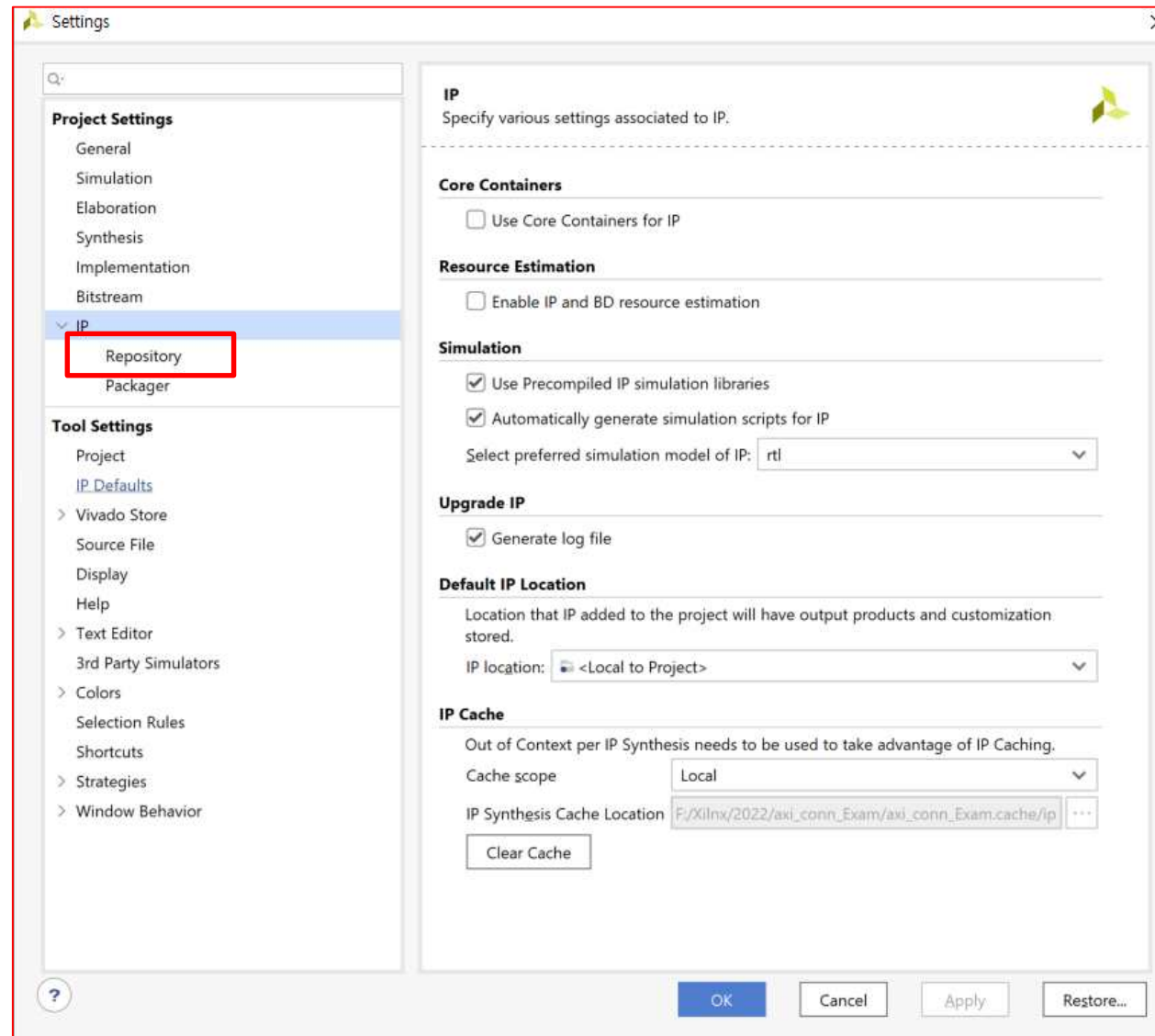
Status: Not started
 Messages: No errors or warnings

Design Runs

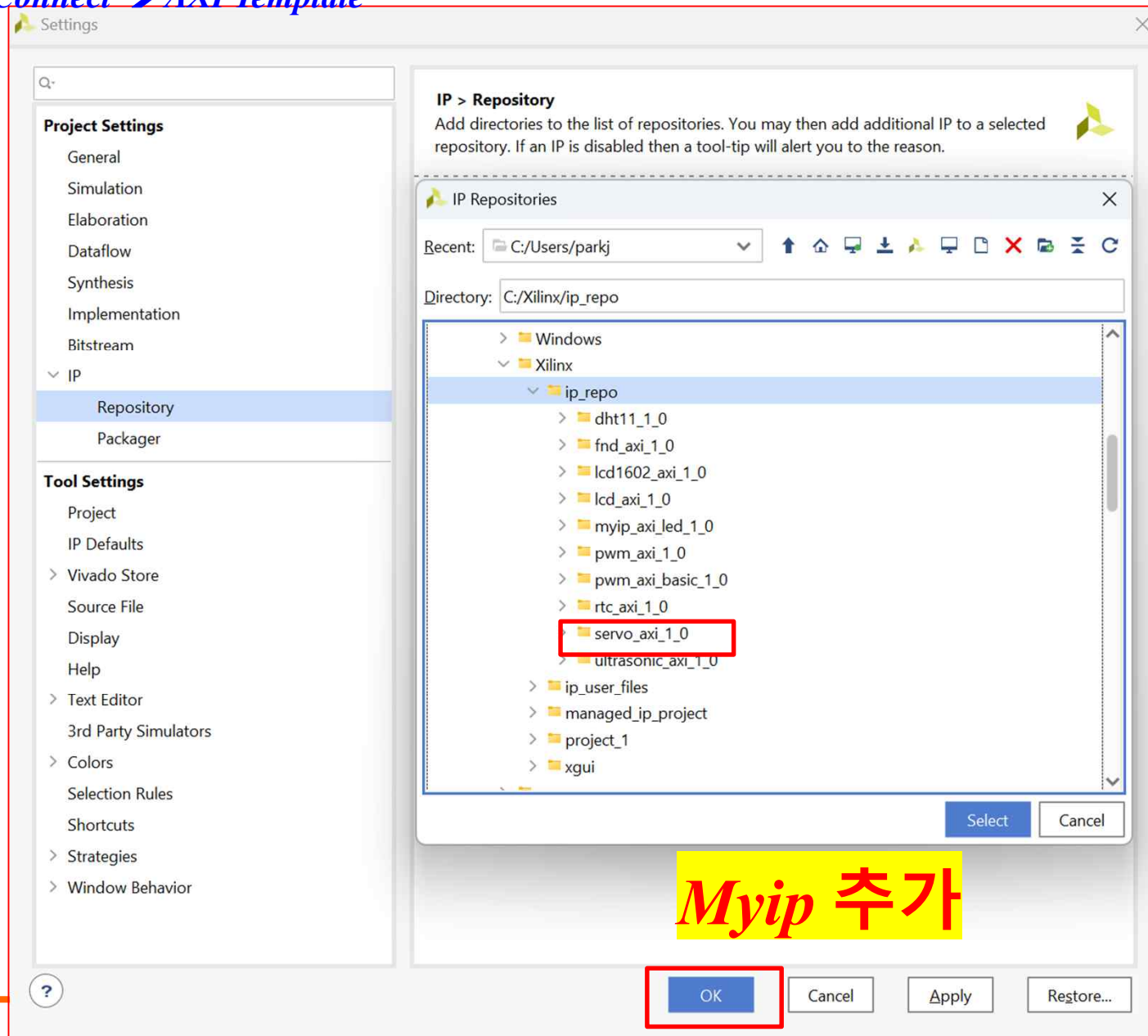
Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started																			Vivado Synthesis Defaults (Viva
impl_1	constrs_1	Not started																			Vivado Implementation Default

➤ AXI Connect ➔ AXI Template

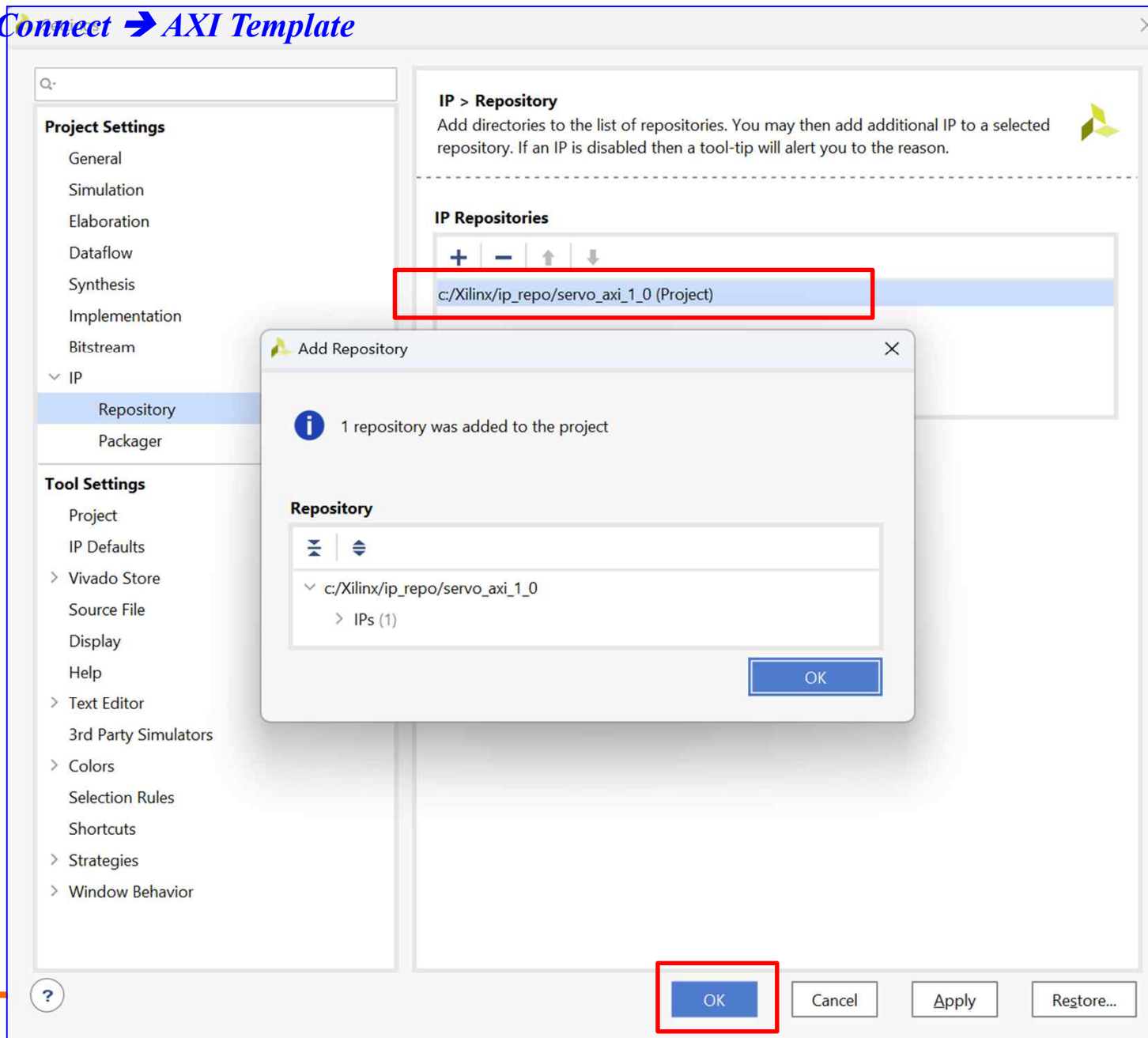
Repository



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

Project Summary x IP Catalog x

Cores | Interfaces

Search:

IP Catalog에서 확인

Name	AXI4	Status	License	VLNV
> User Repository (c:/Xilinx/ip_repo/servo_axi_1_0)				
✓ Vivado Repository				
> Alliance Partners				
> Audio Connectivity & Processing				
> Automotive & Industrial				
> AXI Infrastructure				
> AXIS Infrastructure				
> BaselP				
> Basic Elements				
> Communication & Networking				

➤ AXI Connect ➔ AXI Template

Create Block Design

Project Manager - project_1

Sources

- Design Sources
- Constraints
- Simulation Sources
 - sim_1
- Utility Sources

Repository Properties

User Repository

Path: c:/Xilinx/ip_repo/axi_fnd_1_0

Number of IPs: 1

Number of interfaces: 0

General | IPs

Project Summary | IP Catalog

Overview | Dashboard

Settings | Edit

Project name: project_1
 Project location: C:/Users/parkj/test231013/project_1
 Product family: Artix-7
 Project part: Basys3 (xc7a35tcpg236-1)
 Top module name: Not defined
 Target language: Verilog
 Simulator language: Mixed

Board Part

Display name: Basys3
 Board part name: digilentinc.com:basys3:part0:1.2
 Board revision: C.0
 Connectors: No connections
 Repository path: E:/Xilinx/Vivado/2022.2/data/boards/board_files
 URL: <https://digilent.com/reference/programmable-logic/basys-3/start>
 Board overview: Basys3

Synthesis | **Implementation**

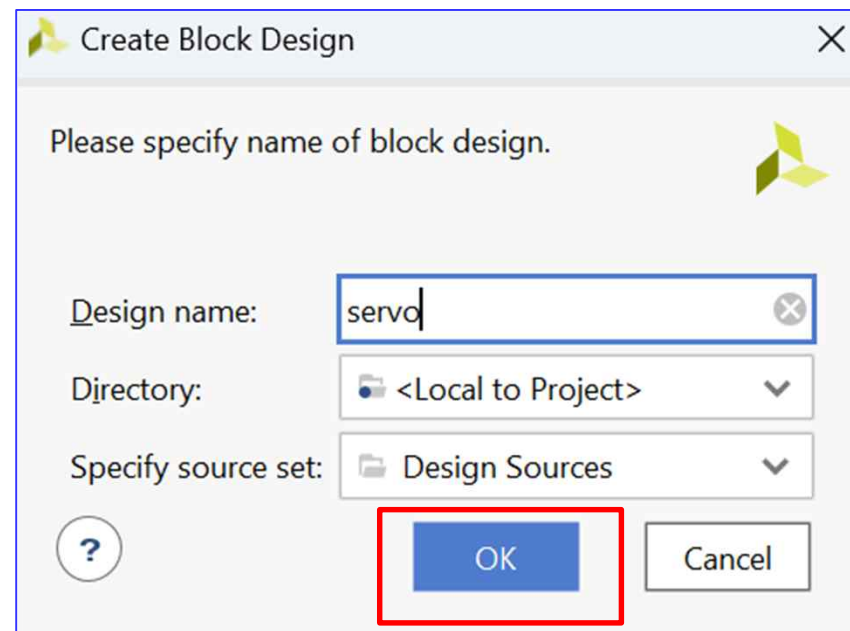
Status: Not started
 Messages: No errors or warnings

Tcl Console | Messages | Log | Reports | **Design Runs**

Name	Constraints	Status	WNS	TNS	WHS	THS	WBSS	TPWS	Total Power	Failed Routes	Methodology	RQA Score	QoR Suggestions	LUT	FF	BRAM	URAM	DSP	Start	Elapsed	Run Strategy
synth_1	constrs_1	Not started																			Vivado Synthesis Defaults (Viva
impl_1	constrs_1	Not started																			Vivado Implementation Default

Create and add an IP subsystem to the project

➤ AXI Connect ➔ AXI Template



이름 설정 후 **OK**

➤ AXI Connect ➔ AXI Template

Microblaze, uart, gpio, servo_axi 추가

Search: microblaze

- MicroBlaze
- MicroBlaze Debug Module (MDM)
- MicroBlaze MCS

Search: uart

- AXI UART16550
- AXI Uartlite

Search: gpio

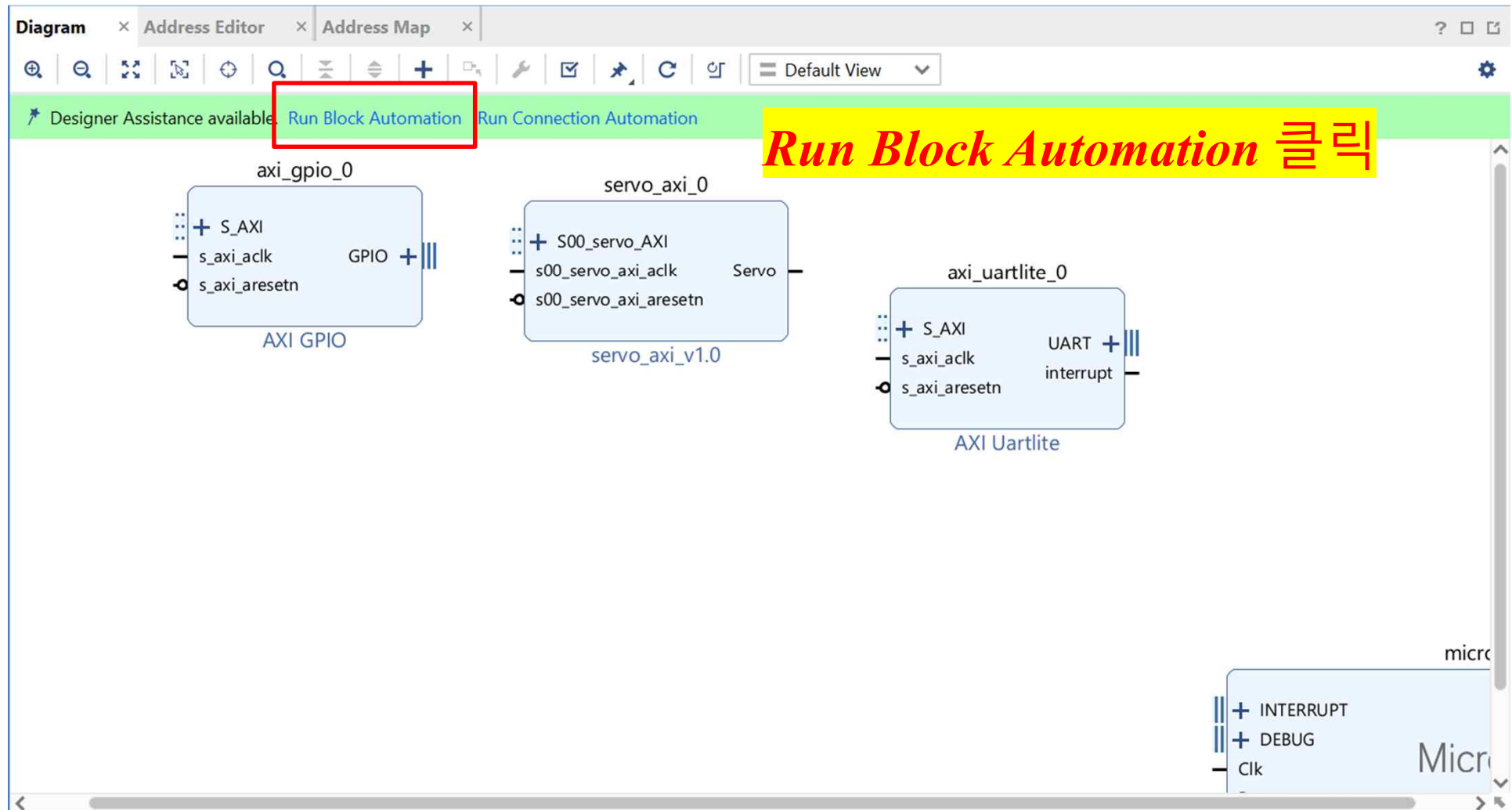
- AXI GPIO

Search: servo

- servo_axi_v1.0

ENTER to select, ESC to cancel, Ctrl+Q for IP data

➤ AXI Connect ➔ AXI Template



AMBA and AXI Connect [Servo Motor]

Axi_Servo_Exam.xpr

Run Block Automation



➤ AXI Connect ➔ AXI Template

Automatically make connections in your design by checking the boxes of the blocks to connect. Select a block on the left to display its configuration options on the right.



▼ ☒ All Automation (1 out of 1 selected)

☒ microblaze_0

Description

MicroBlaze connection automation generates local memory of selected size, and caches can be configured. MicroBlaze Debug Module, Peripheral AXI Interconnect, Interrupt Controller, a clock source, Processor System Reset are added and connected as needed. A preset MicroBlaze configuration can also be selected.

Information about the options can be found in the tooltips.

Options

Preset None ▼

Local Memory 32KB ▼

Local Memory ECC None ▼

Cache Configuration None ▼

Debug Module Debug Only ▼

Peripheral AXI Port Enabled ▼

☐ Interrupt Controller

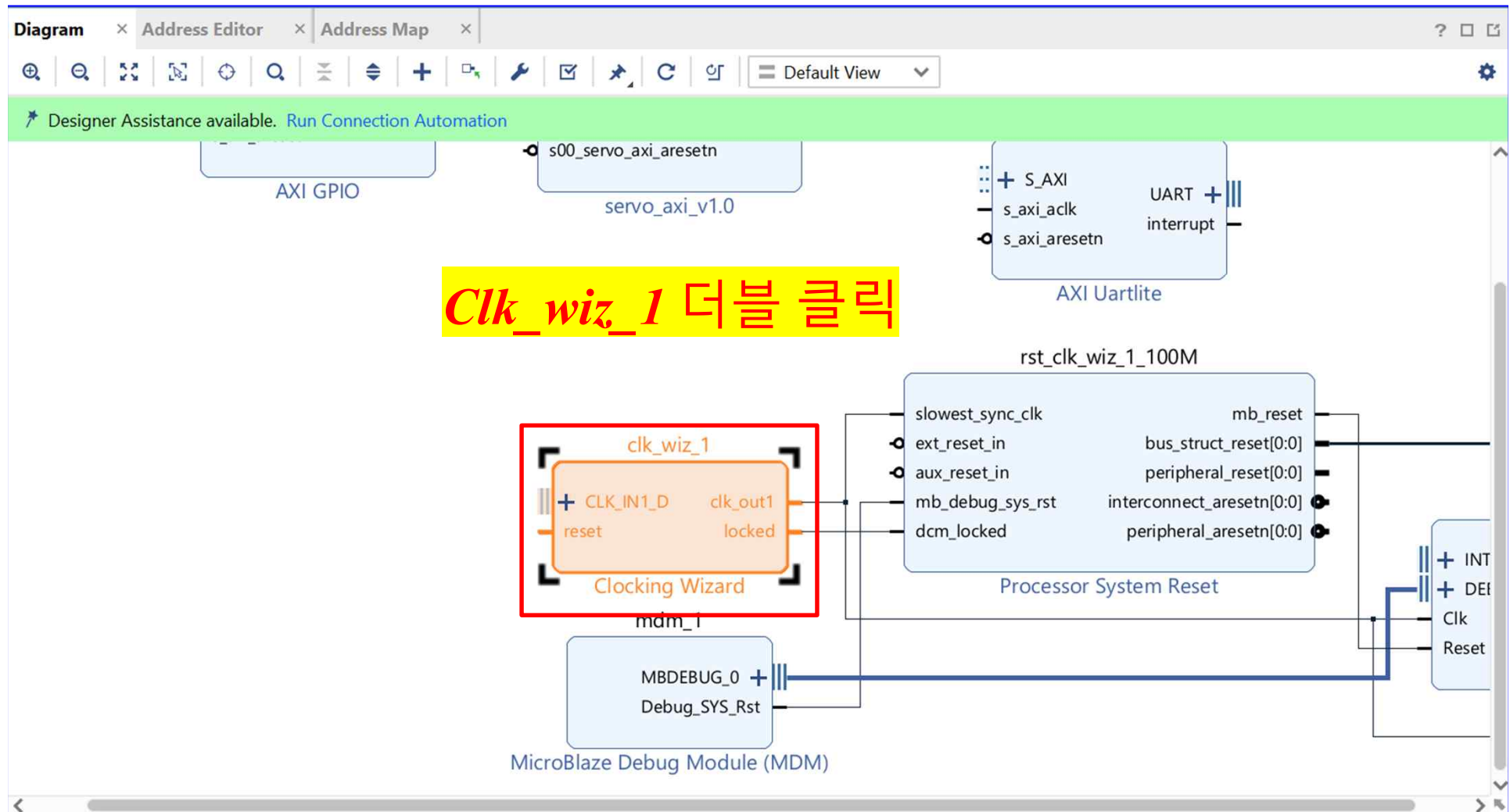
Clock Connection New Clocking Wizard ▼



OK

Cancel

➤ AXI Connect ➔ AXI Template



Re-customize IP

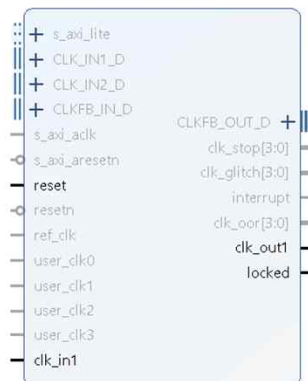
AXI Connect → AXI Template

Documentation IP Location

IP Symbol

Resource

☒ Show disabled ports



Component Name clk_wiz_1

Board

Clocking Options

Output Clocks

MMCM Settings

Summary

Clock Monitor

☐ Enable Clock Monitoring

Primitive

☒ MMCM ☐ PLL

Clocking Features

☒ Frequency Synthesis ☐ Minimize Power
☒ Phase Alignment ☐ Spread Spectrum
☐ Dynamic Reconfig ☐ Dynamic Phase Shift
☐ Safe Clock Startup

Jitter Optimization

☒ Balanced
☐ Minimize Output Jitter
☐ Maximize Input Jitter filtering

Dynamic Reconfig Interface

Options

☒ AXI4Lite ☐ DRP ☐ Phase Duty Cycle Config ☐ Write DRP registers

Input Clock Information

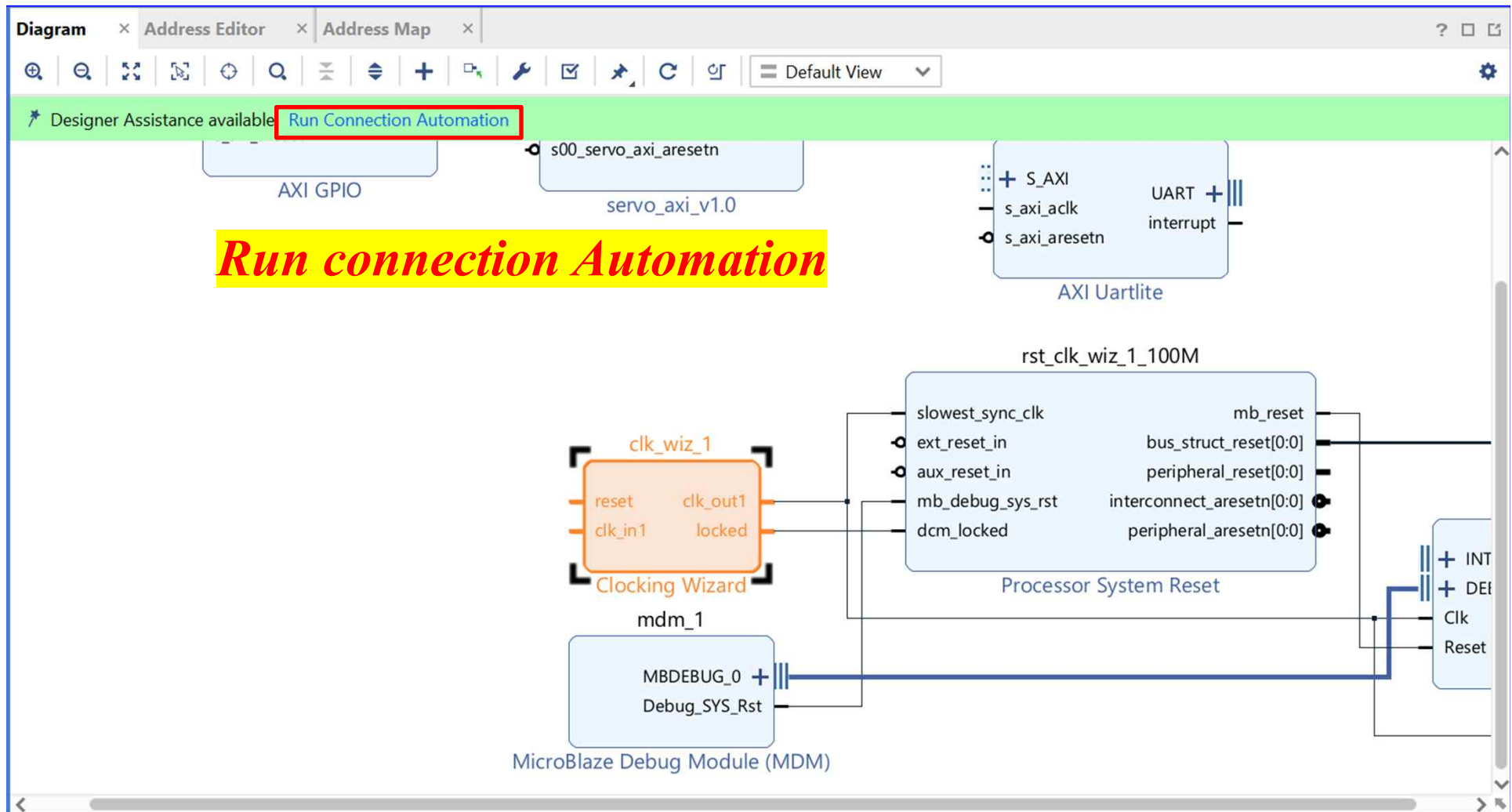
	Input Clock	Port Name	Input Frequency(MHz)		Jitter Options	Input Jitter	Source
☒	Primary	clk_in1	100.000	10.000 - 800.000	UI	0.010	Single ended clock capable p
☐	Secondary	clk_in2	100.000	60.000 - 120.000		0.010	Single ended clock capable p

Single ended clock 변경 후 OK

OK

Cancel

➤ AXI Connect ➔ AXI Template



☒ All Automation (8 out of 8 selected)

- ☒ axi_gpio_0
 - ☒ GPIO
 - ☒ S_AXI
- ☒ axi_uartlite_0
 - ☒ S_AXI
 - ☒ UART
- ☒ clk_wiz_1
 - ☒ clk_in1
 - ☒ reset
- ☒ rst_clk_wiz_1_100M
 - ☒ ext_reset_in
- ☒ servo_axi_0
 - ☒ S00_servo_AXI

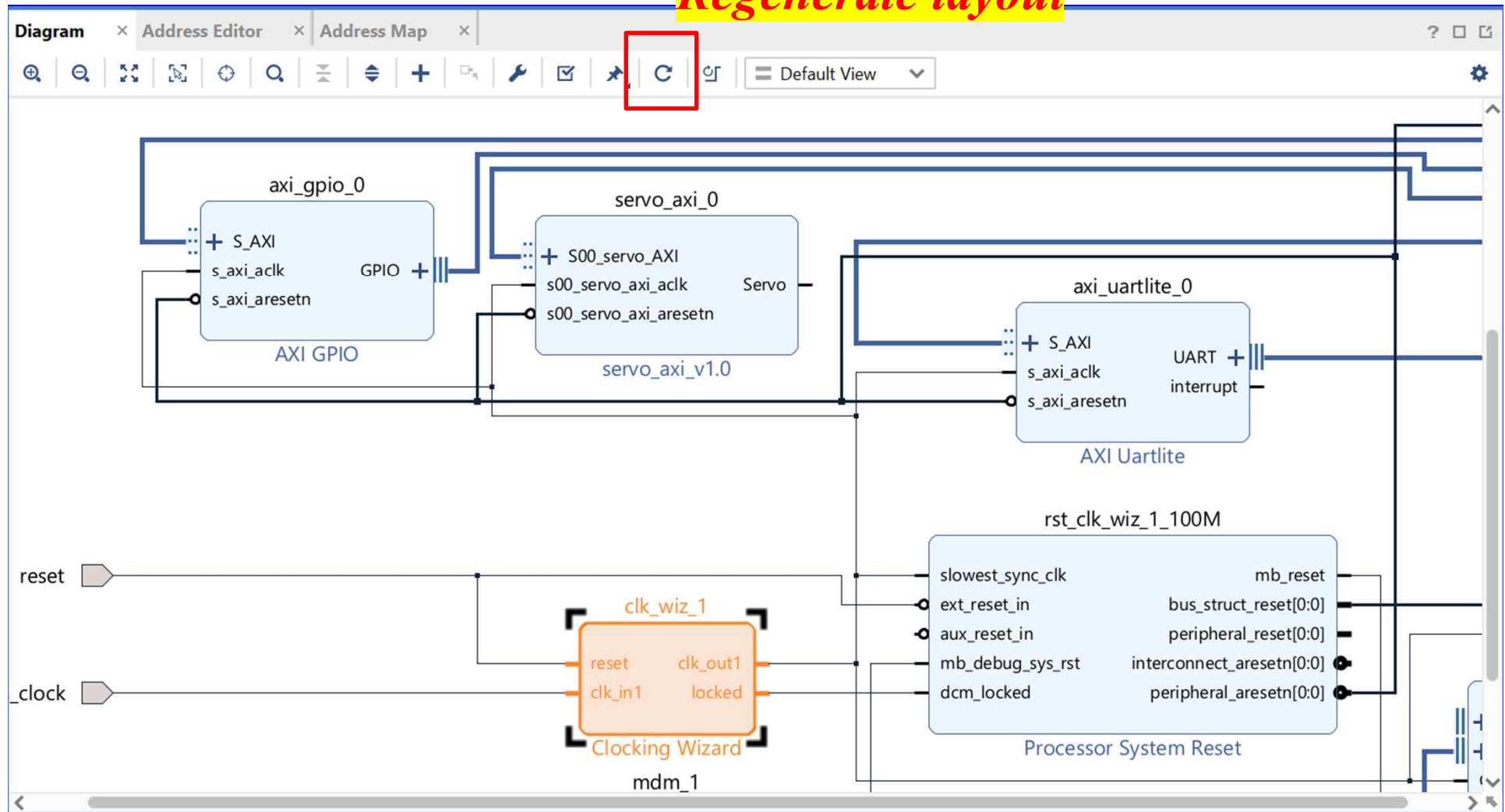
Select an interface pin on the left panel to view its options

All Automation 선택 후 OK

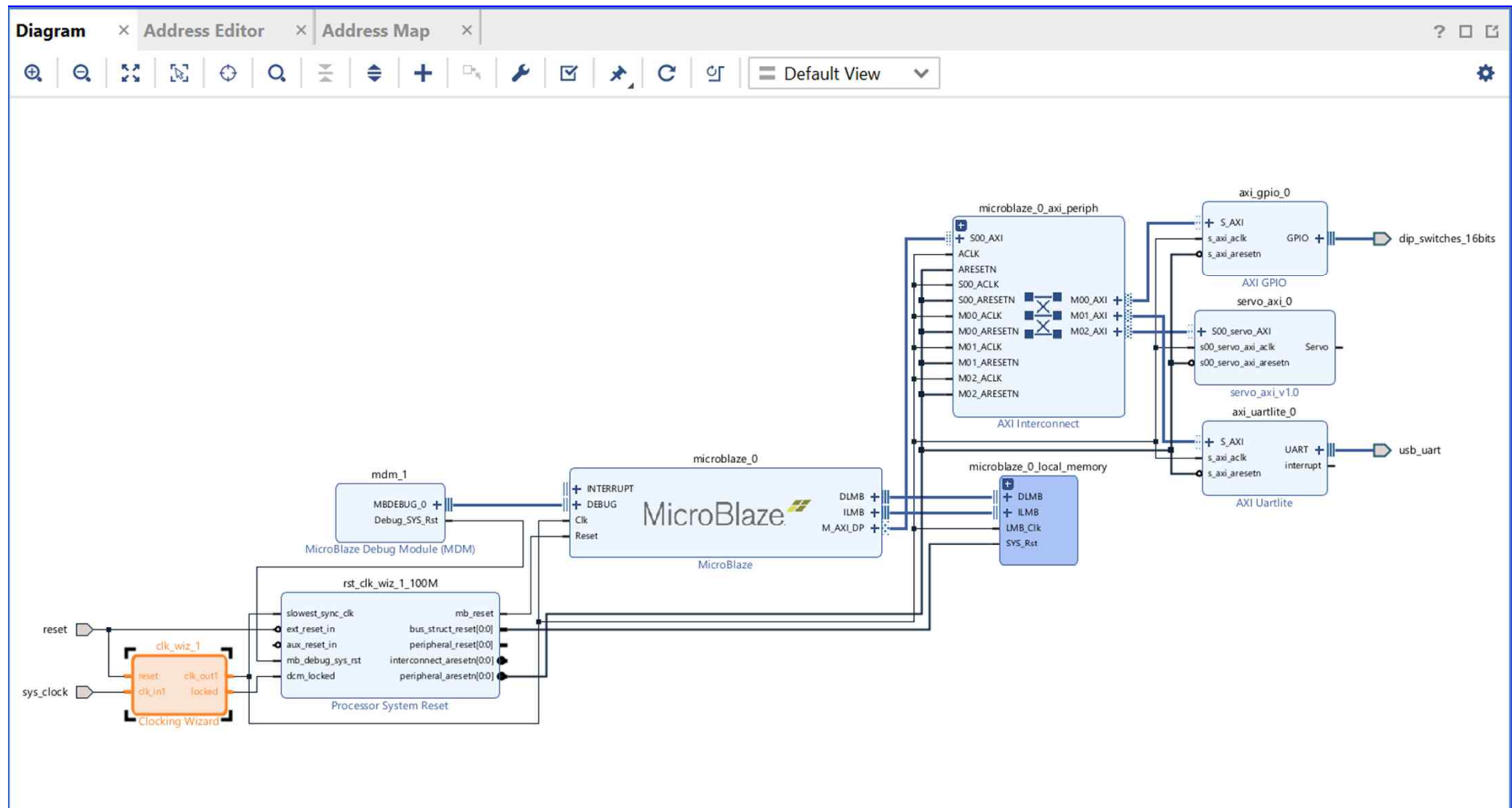
OK Cancel

➤ AXI Connect ➔ AXI Template

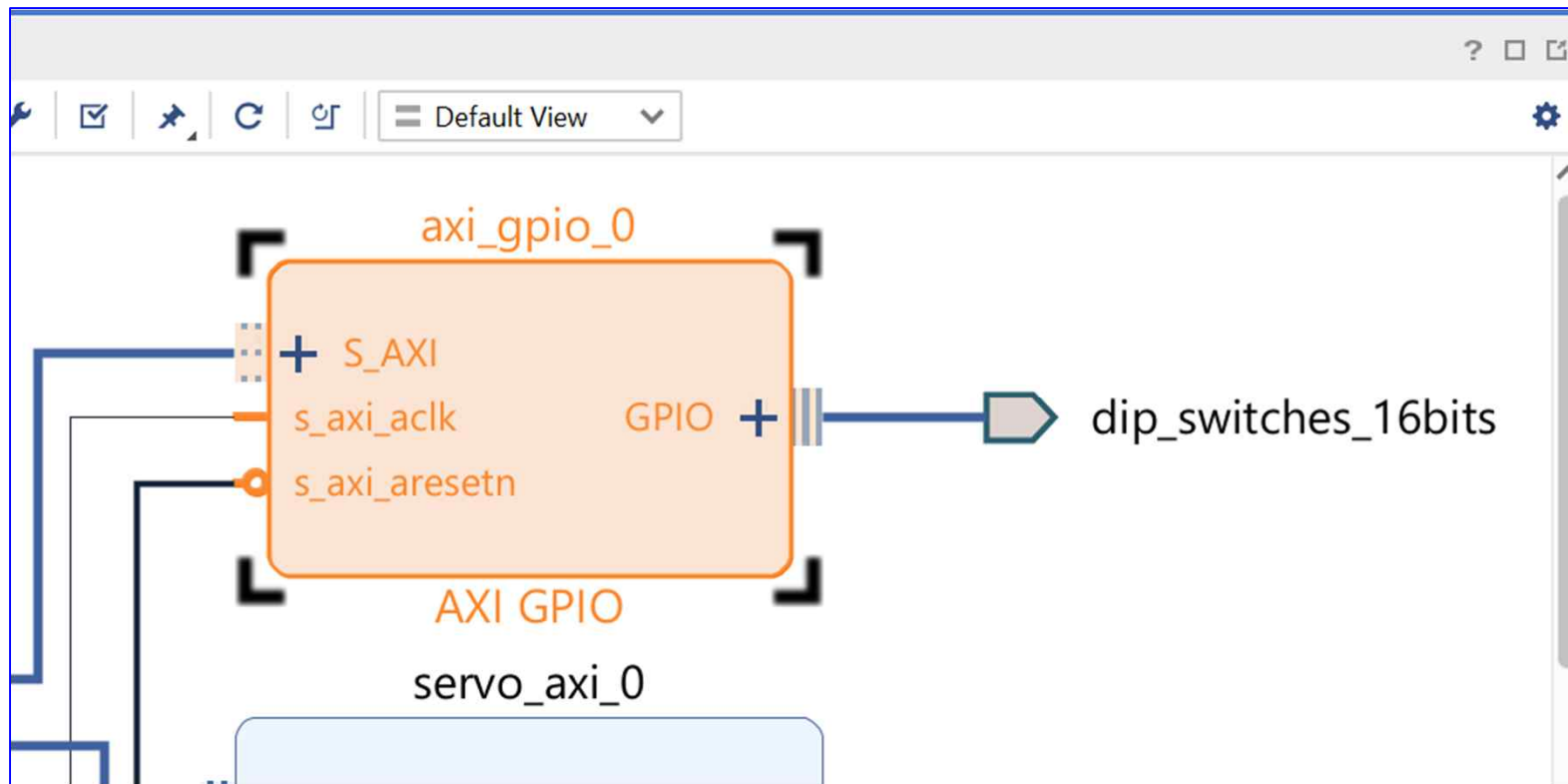
Regenerate layout



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



GPIO IP 수정 ➔ Next

➤ AXI Connect ➔ AXI Template

AXI GPIO (2.0)

Documentation IP Location

☒ Show disabled ports

Component Name axi_gpio_0

Board **IP Configuration**

Associate IP interface with board interface

IP Interface	Board Interface
GPIO	Custom
GPIO2	Custom

Clear Board Parameters

☐ Enable Interrupt

OK Cancel

GPIO IP 수정 ➔ Next

AMBA and AXI Connect [Servo Motor]

➤ AXI GPIO (2.0) → AXI Template

GPIO IP 수정 → Next

Documentation IP Location

☒ Show disabled ports



Component Name axi_gpio_0

Board IP Configuration

GPIO

☒ All Inputs

☐ All Outputs

GPIO Width 4 [1 - 32]

Default Output Value 0x00000000 [0x00000000,0xFFFFFFFF]

Default Tri State Value 0xFFFFFFFF [0x00000000,0xFFFFFFFF]

☐ Enable Dual Channel

GPIO 2

☐ All Inputs

☐ All Outputs

GPIO Width 32 [1 - 32]

Default Output Value 0x00000000 [0x00000000,0xFFFFFFFF]

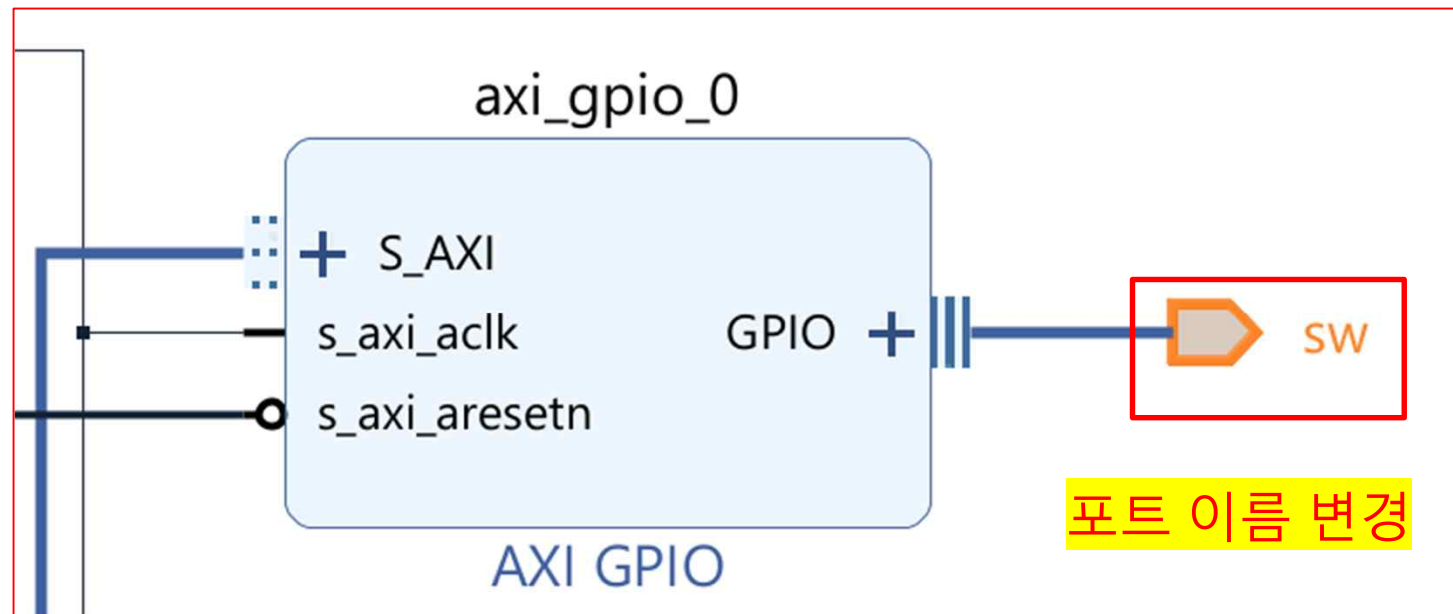
Default Tri State Value 0xFFFFFFFF [0x00000000,0xFFFFFFFF]

☐ Enable Interrupt

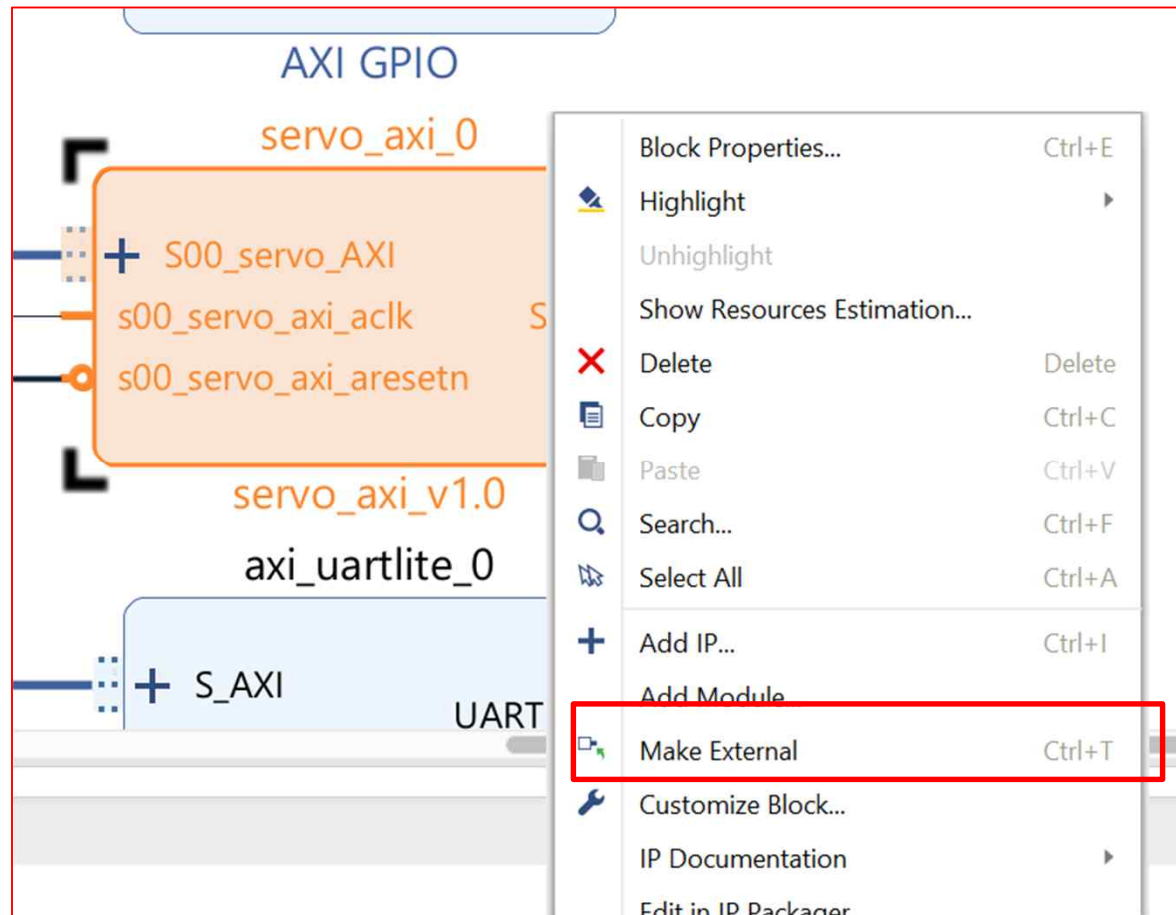
OK

Cancel

➤ AXI Connect ➔ AXI Template

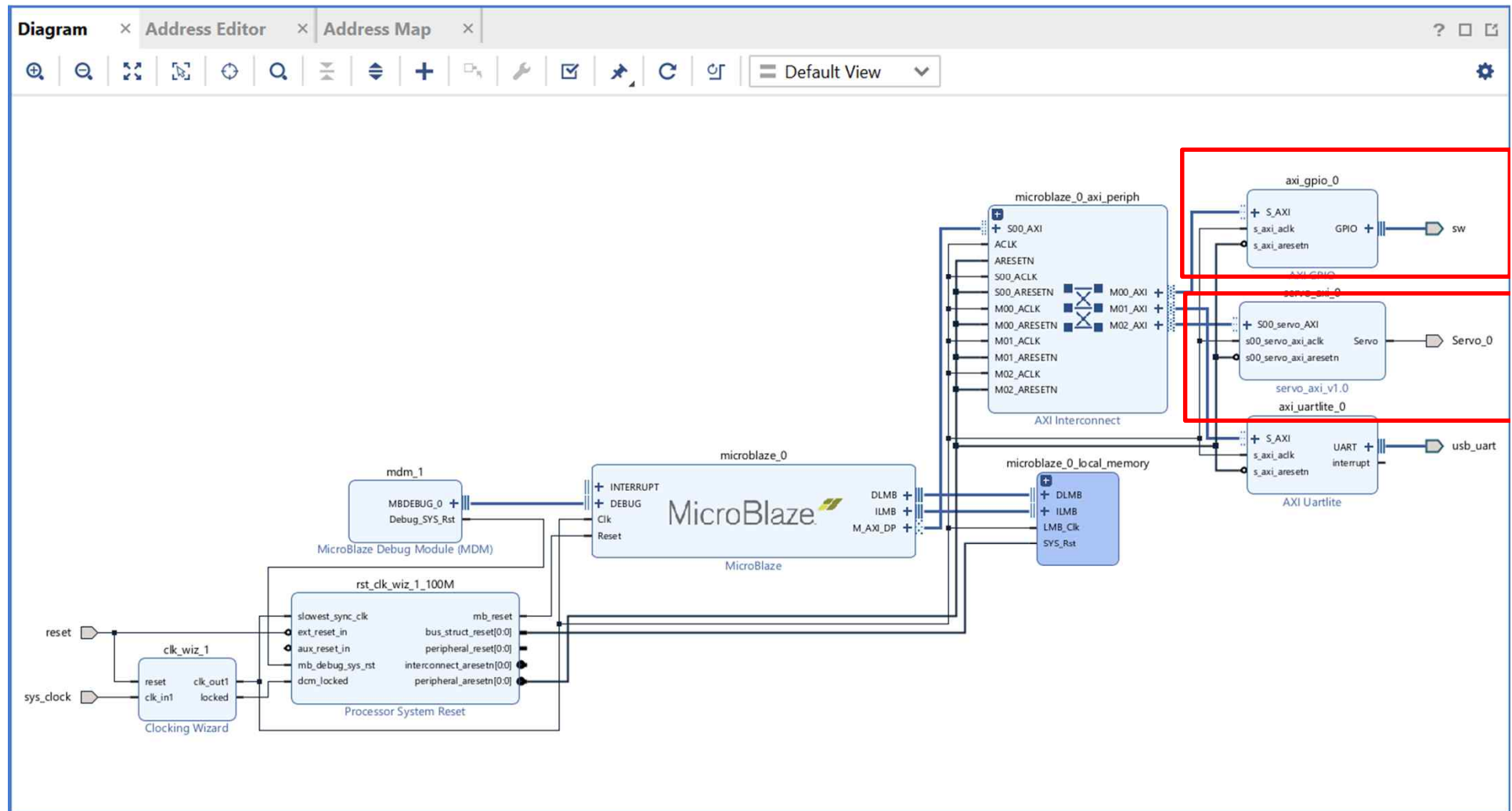


➤ AXI Connect ➔ AXI Template ➔ servo_axi_0 ➔ Make External

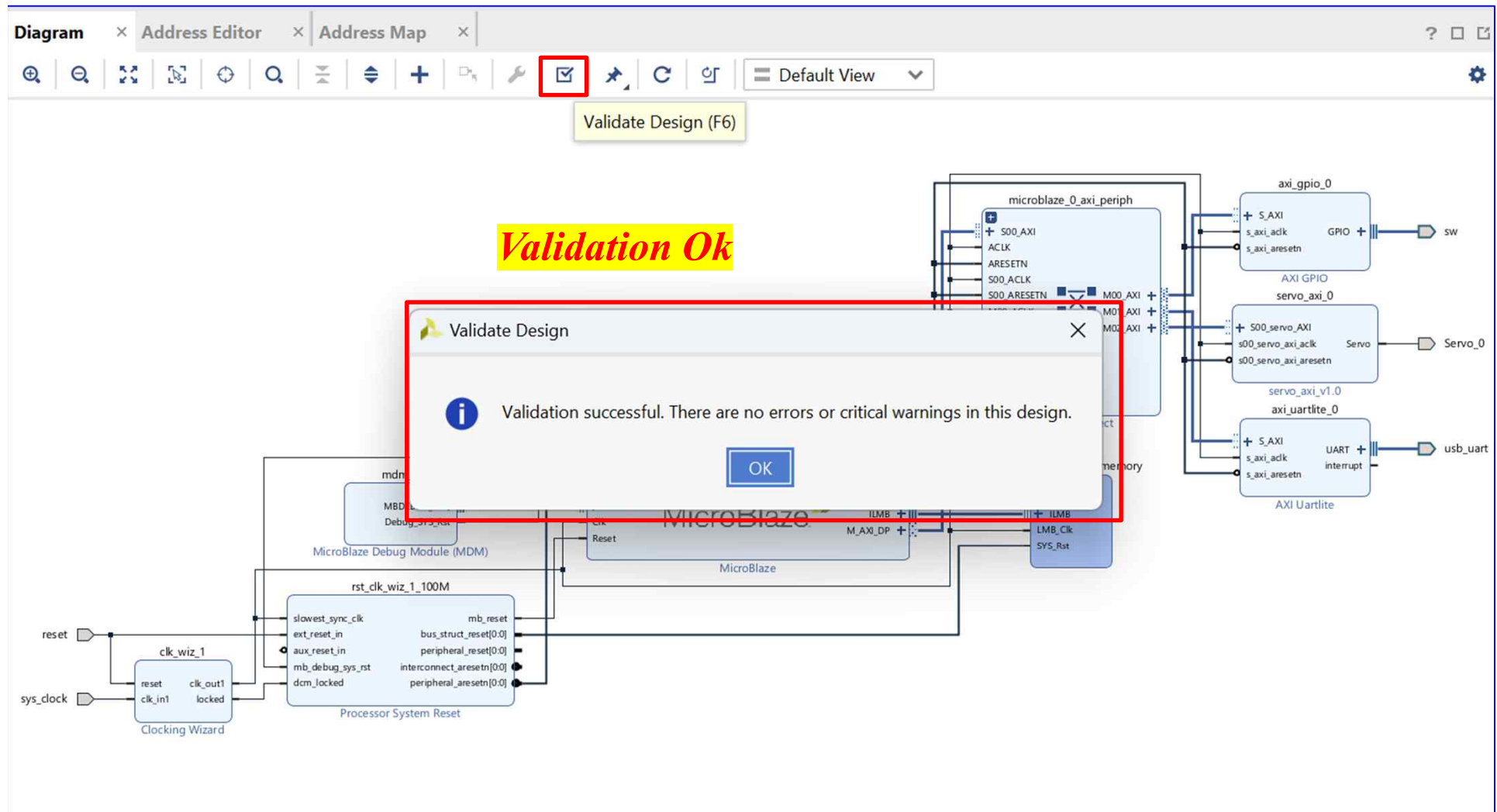


Servo IP make External

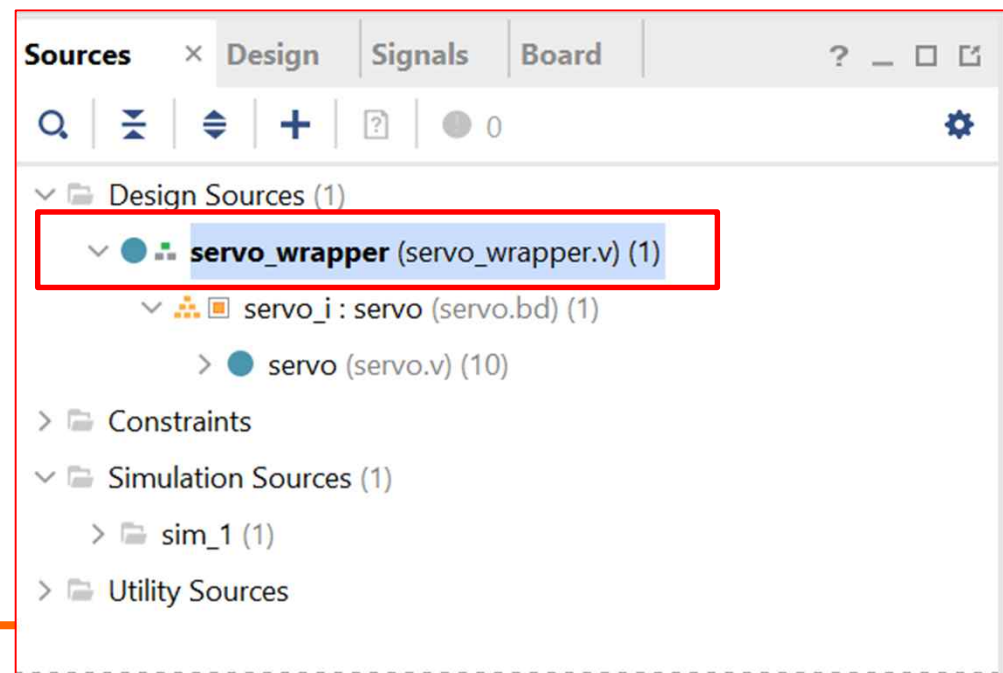
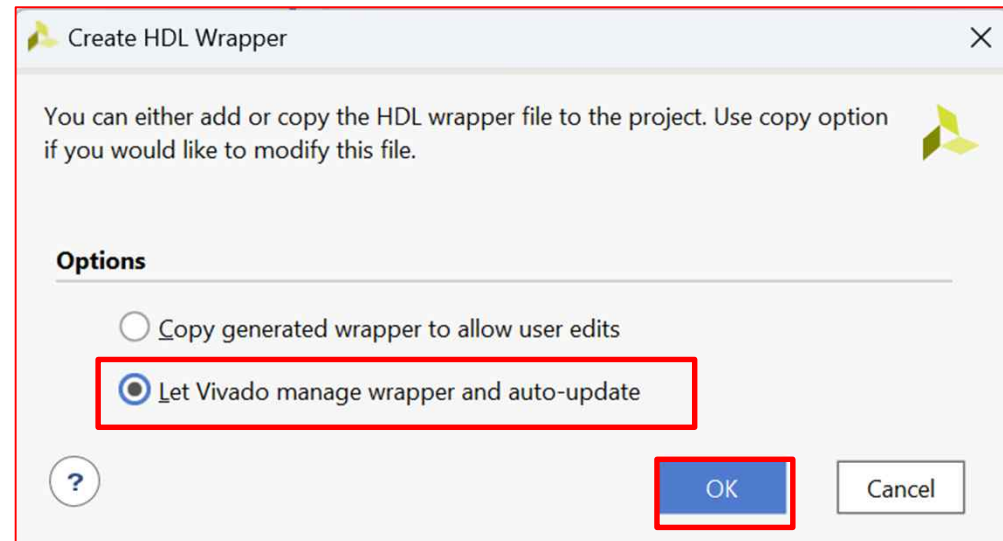
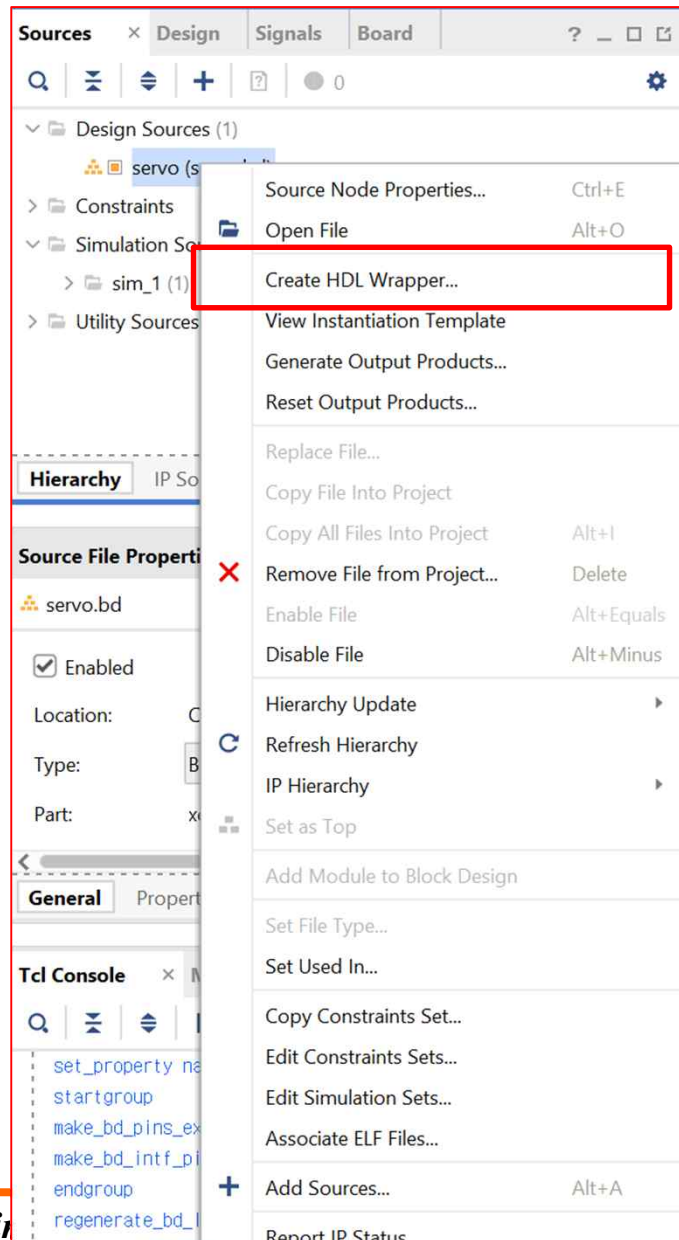
➤ AXI Connect ➔ AXI Template



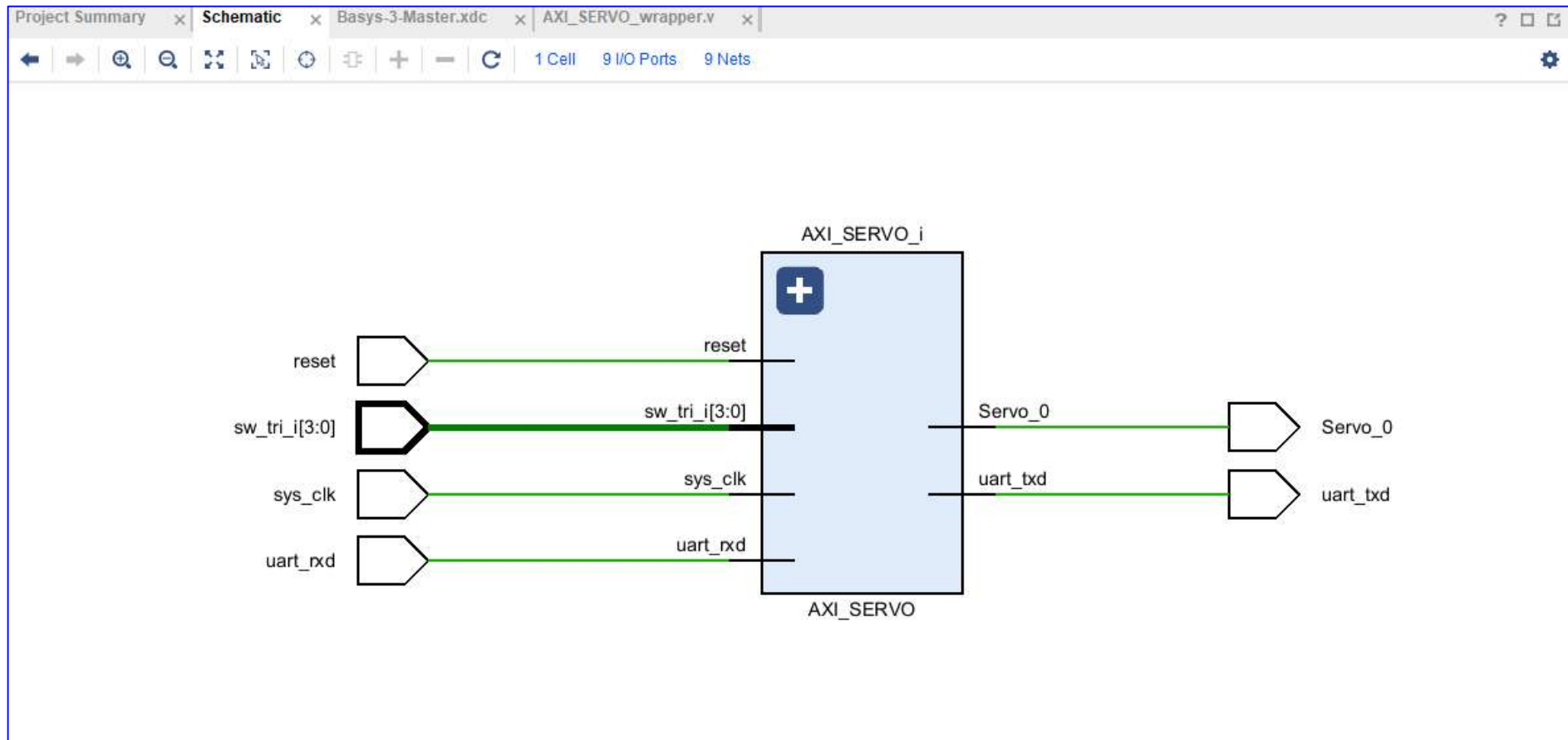
➤ AXI Connect ➔ AXI Template ➔ **Validate**



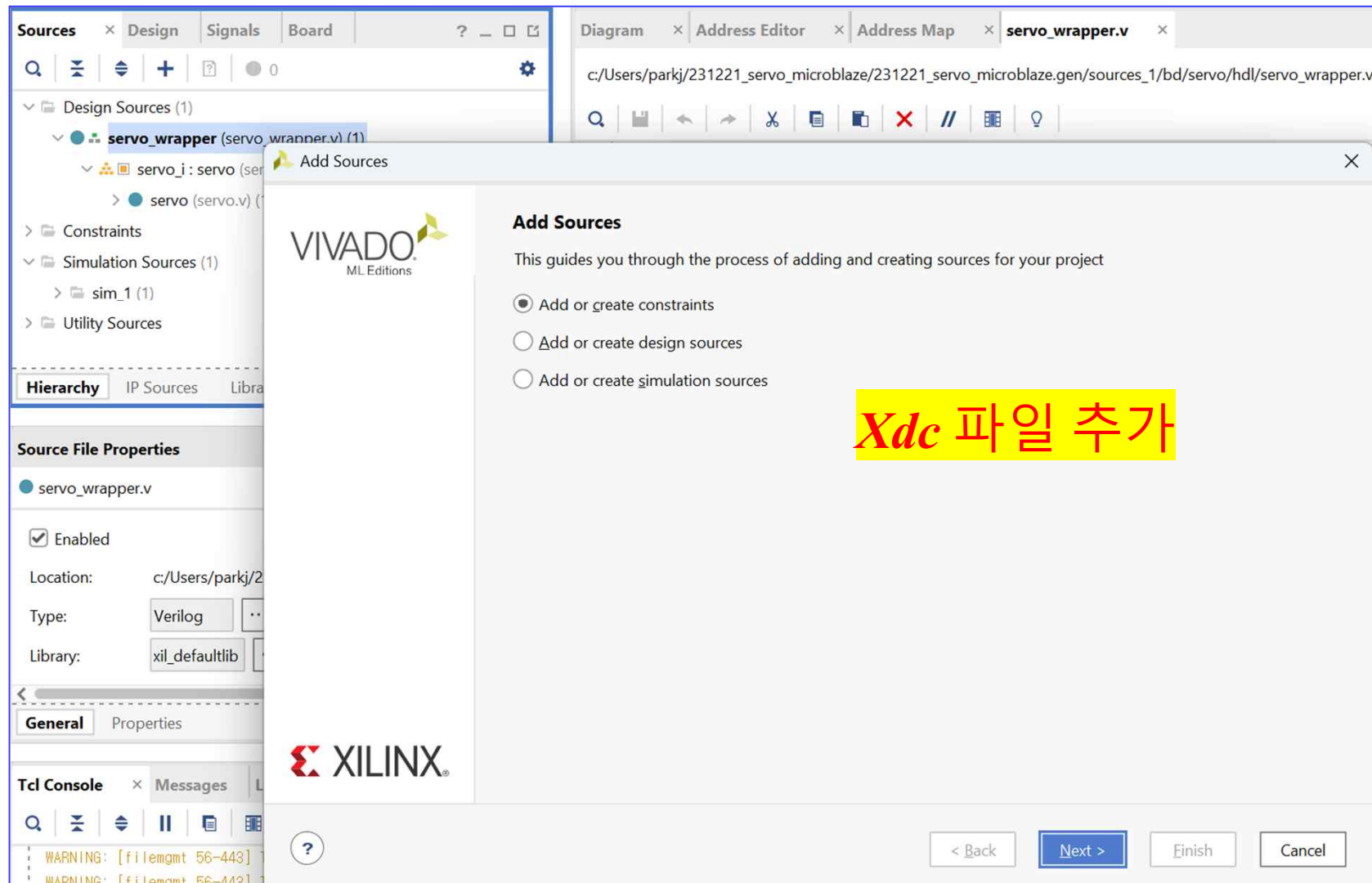
➤ AXI Connect ➔ AXI Template ➔ Create HDL Wrapper



➤ AXI Connect ➔ AXI Template ➔ **RTL** ➔ Schematic



➤ AXI Connect ➔ AXI Template



Basys-3-Master.xdc

➤ AXI Connect ➔ AXI Template

▪ XDC Constraints

1. `set_property -dict { PACKAGE_PIN W5 IOSTANDARD LVCMOS33 } [get_ports sys_clock]`
2. `create_clock -add -name sys_clk_pin -period 10.00 -waveform {0 5} [get_ports sys_clock]`
3. **## Switches**
4. `set_property -dict { PACKAGE_PIN R2 IOSTANDARD LVCMOS33 } [get_ports {reset}]`
5. `set_property -dict { PACKAGE_PIN V17 IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[0]}]`
6. `set_property -dict { PACKAGE_PIN V16 IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[1]}]`
7. `set_property -dict { PACKAGE_PIN W16 IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[2]}]`
8. `set_property -dict { PACKAGE_PIN W17 IOSTANDARD LVCMOS33 } [get_ports {sw_tri_i[3]}]`
9. **##Pmod Header JC**
10. `set_property -dict { PACKAGE_PIN K17 IOSTANDARD LVCMOS33 } [get_ports {Servo_0}];#Sch
name = JC1`
11. **##USB-RS232 Interface**
12. `set_property -dict { PACKAGE_PIN B18 IOSTANDARD LVCMOS33 } [get_ports usb_uart_rxd]`
13. `set_property -dict { PACKAGE_PIN A18 IOSTANDARD LVCMOS33 } [get_ports usb_uart_txd]`
14. **## Configuration options, can be used for all designs**
15. `set_property CONFIG_VOLTAGE 3.3 [current_design]`
16. `set_property CFGBVS VCCO [current_design]`
17. **## SPI configuration mode options for QSPI boot, can be used for all designs**
18. `set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]`
19. `set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]`
20. `set_property CONFIG_MODE SPIx4 [current_design]`
21. **## Configuration options, can be used for all designs**
22. `set_property CONFIG_VOLTAGE 3.3 [current_design]`
23. `set_property CFGBVS VCCO [current_design]`
24. **## SPI configuration mode options for QSPI boot, can be used for all designs**
25. `set_property BITSTREAM.GENERAL.COMPRESS TRUE [current_design]`
26. `set_property BITSTREAM.CONFIG.CONFIGRATE 33 [current_design]`
27. `set_property CONFIG_MODE SPIx4 [current_design]`

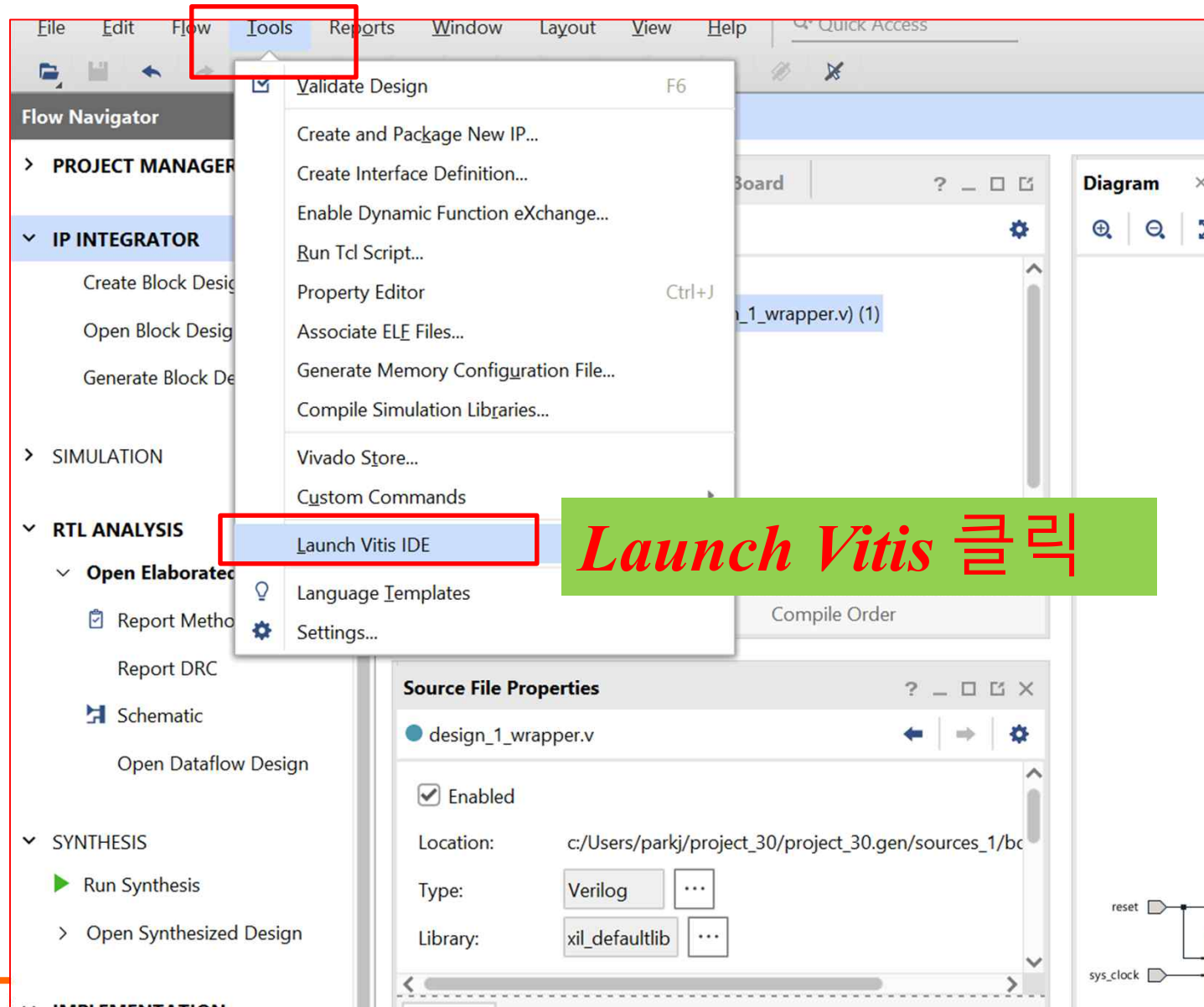
➤ AXI Connect ➔ AXI Template

- Create Block Design
- Blaze Uart & FND 추가 및 연결
- sys_clock & reset 설정
- GPIO, Servo Make External
- XDC Constraints 설정
- **Bitstream**
- **Export Hardwar**

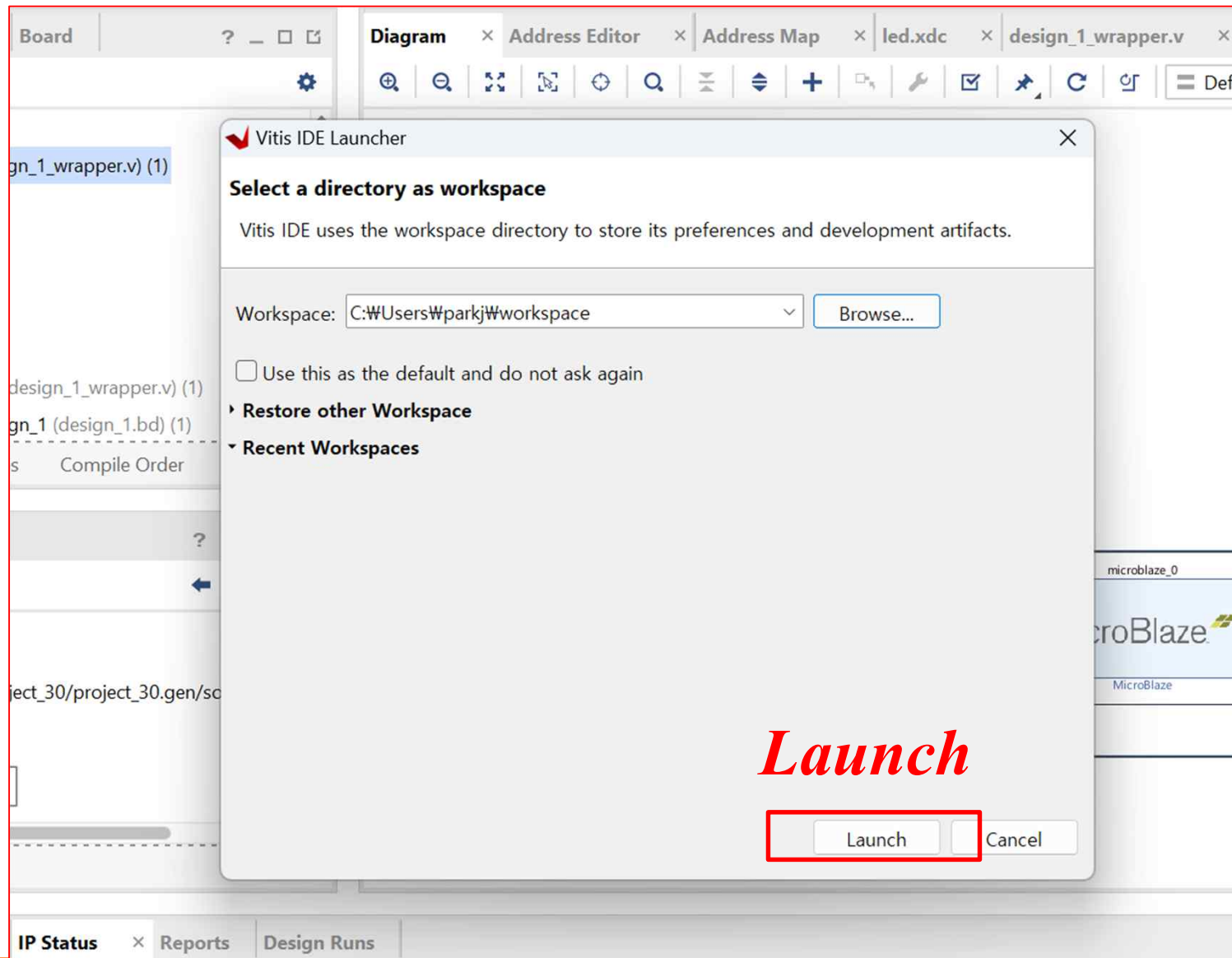
AMBA and AXI Connect [Servo Motor]

✓ *Application SW(Vitis)*

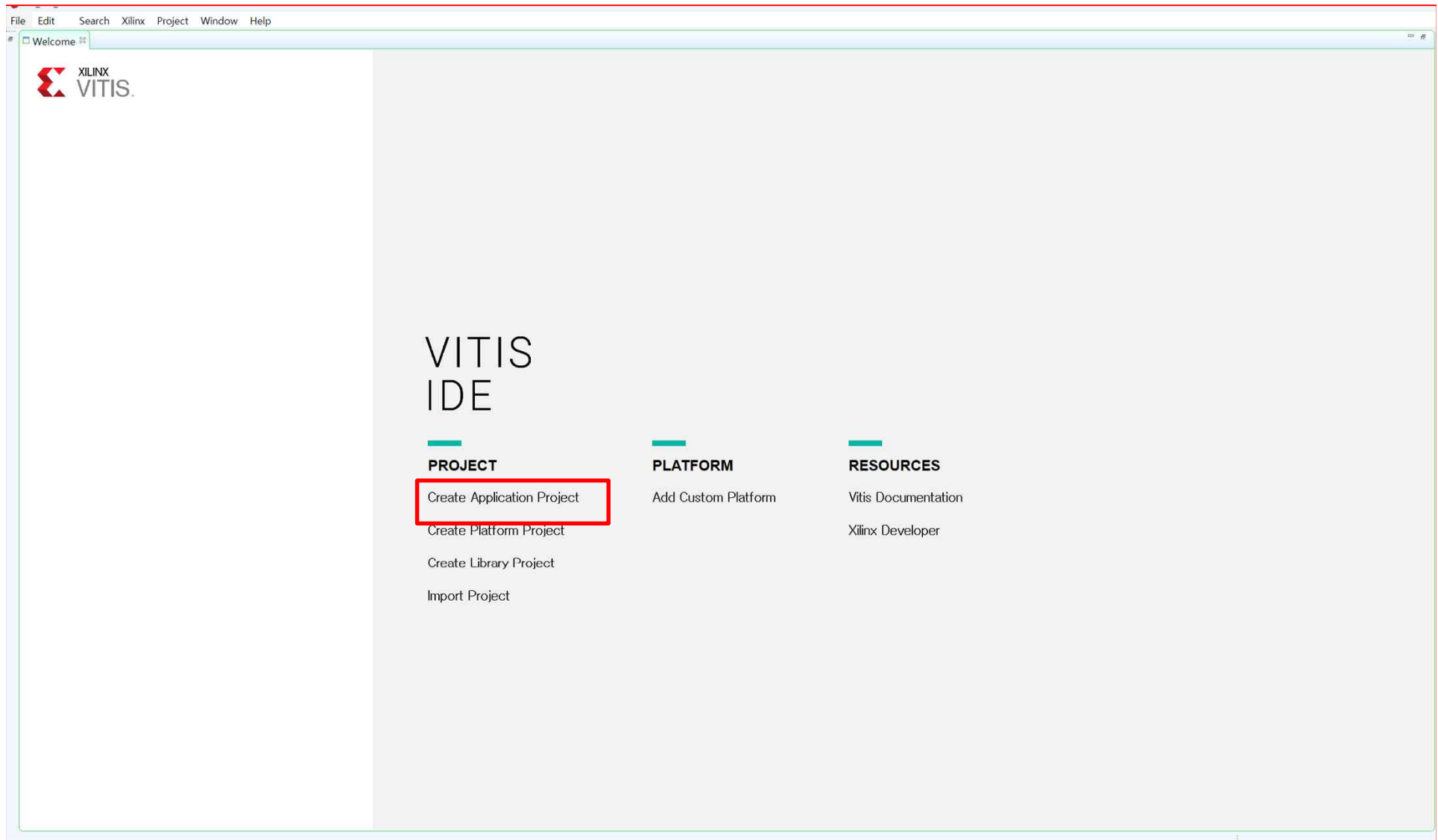
➤ AXI Connect ➔ AXI Template



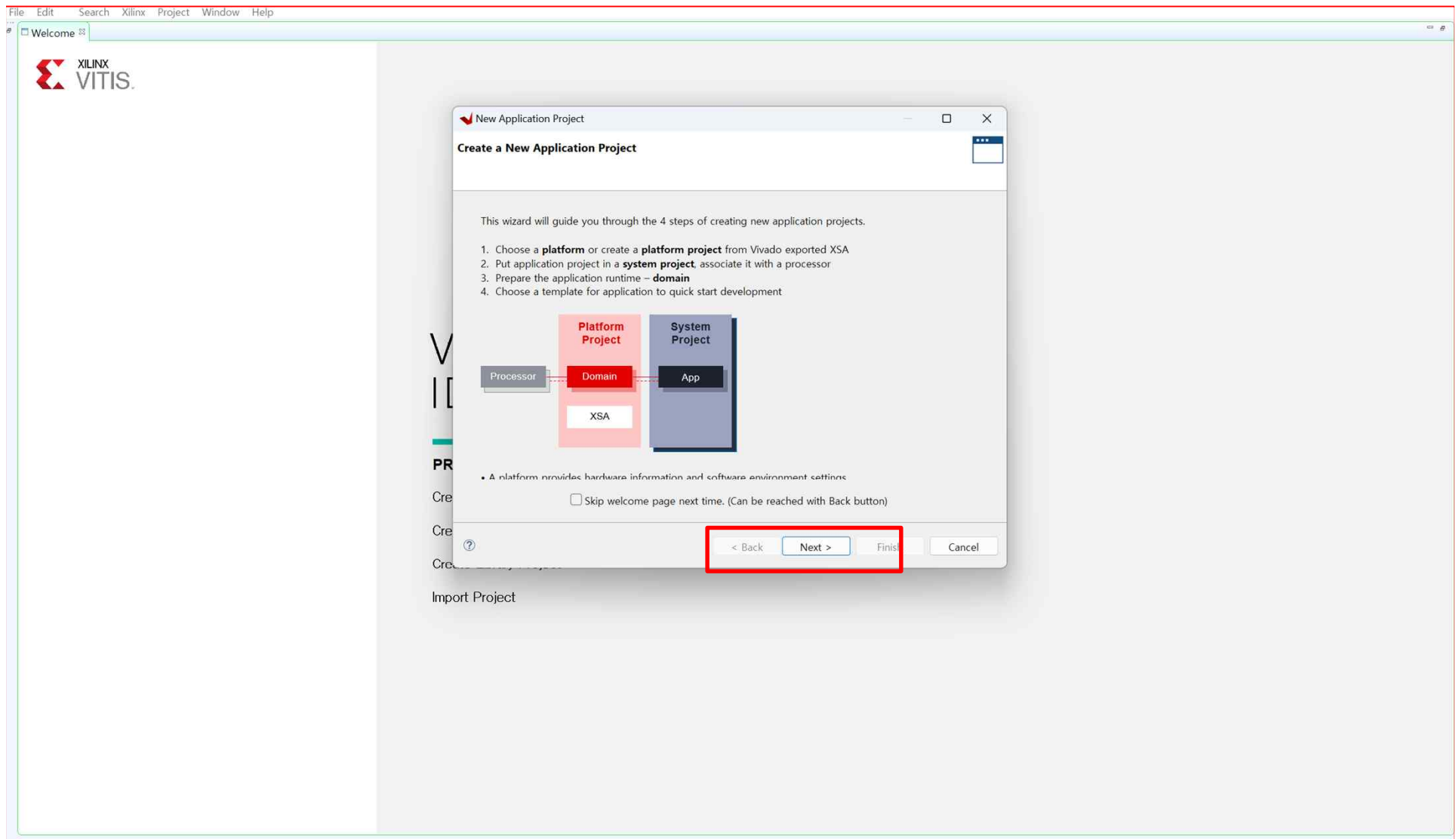
➤ AXI Connect ➔ AXI Template



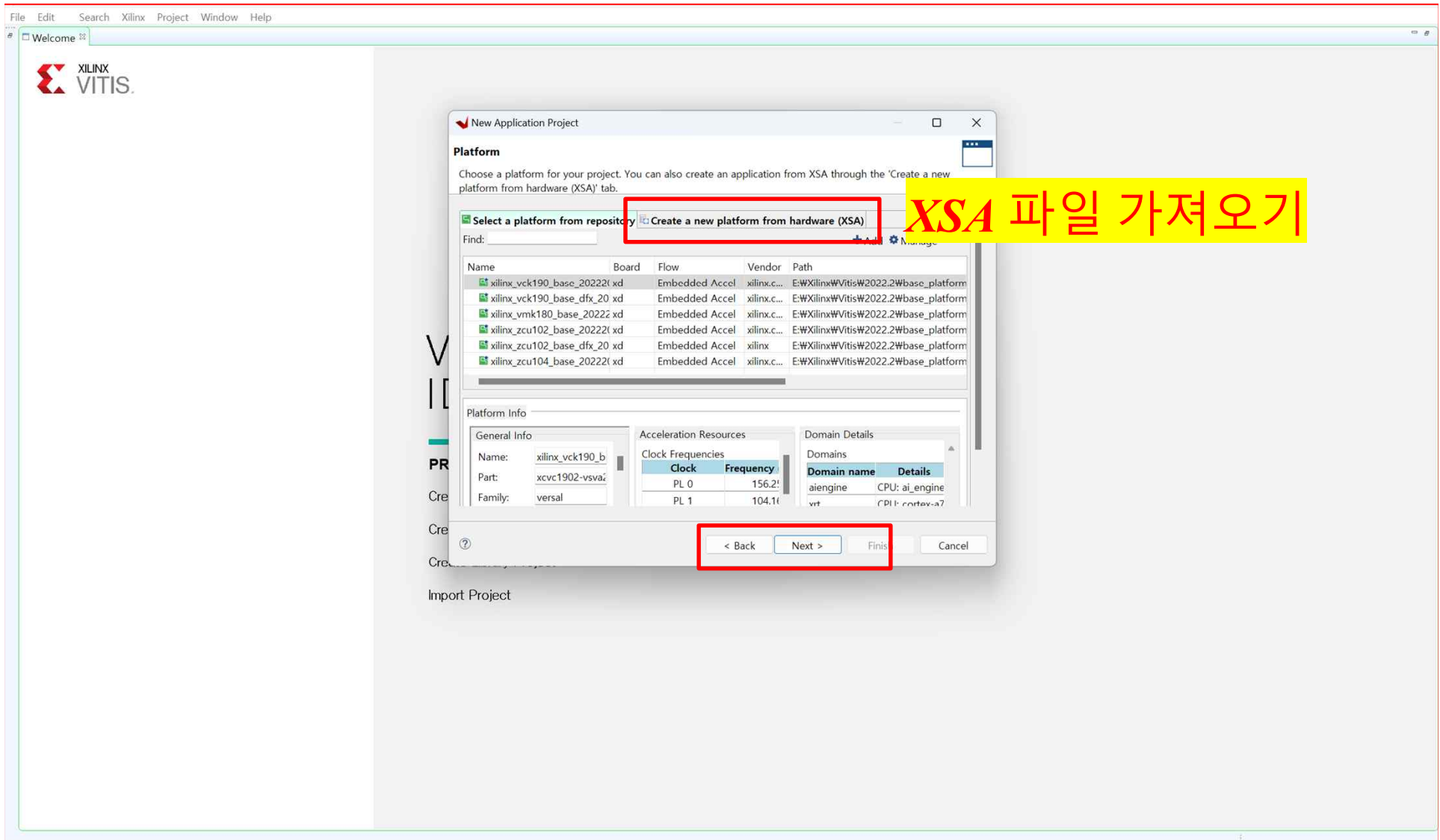
➤ AXI Connect ➔ AXI Template



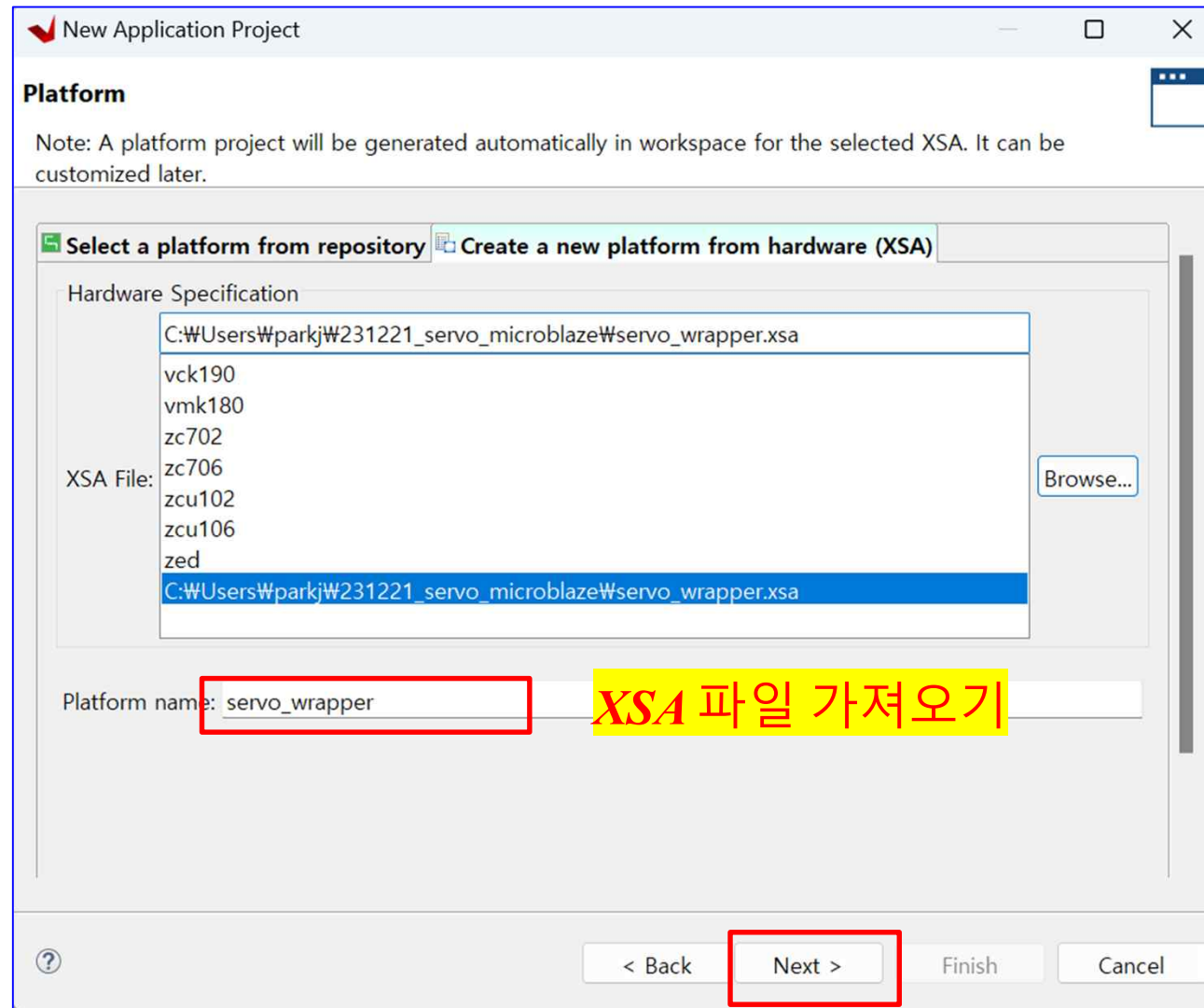
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template

New Application Project

Application Project Details
Specify the application project name and its system project properties

Application project name:

System Project
Create a new system project for the application or select an existing one from the workspace

Select a system project
[+ Create new...](#)

System project details

System project name:

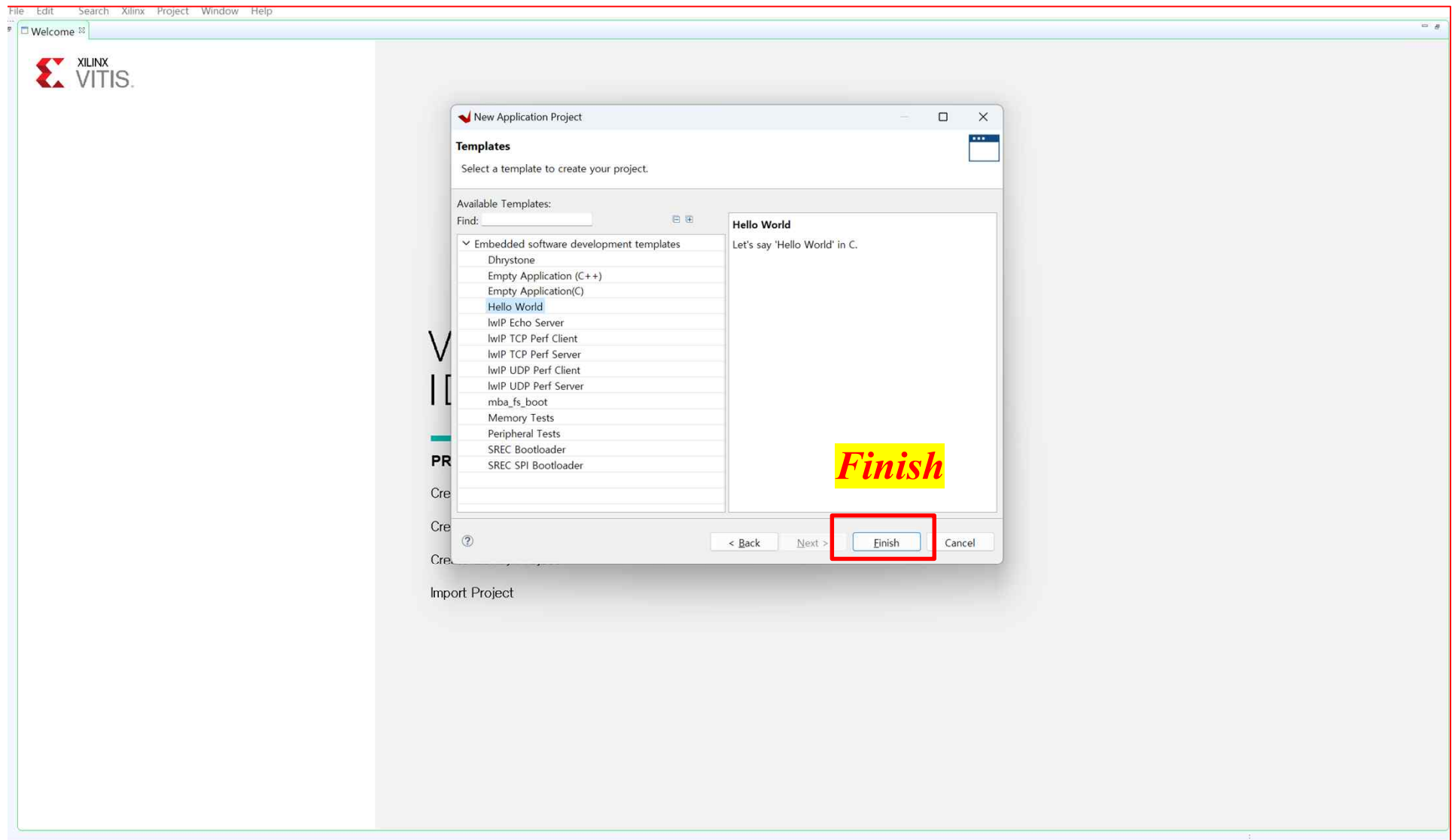
Target processor
Select target processor for the Application project.

Processor	Associated applications	
microblaze_0	servo	

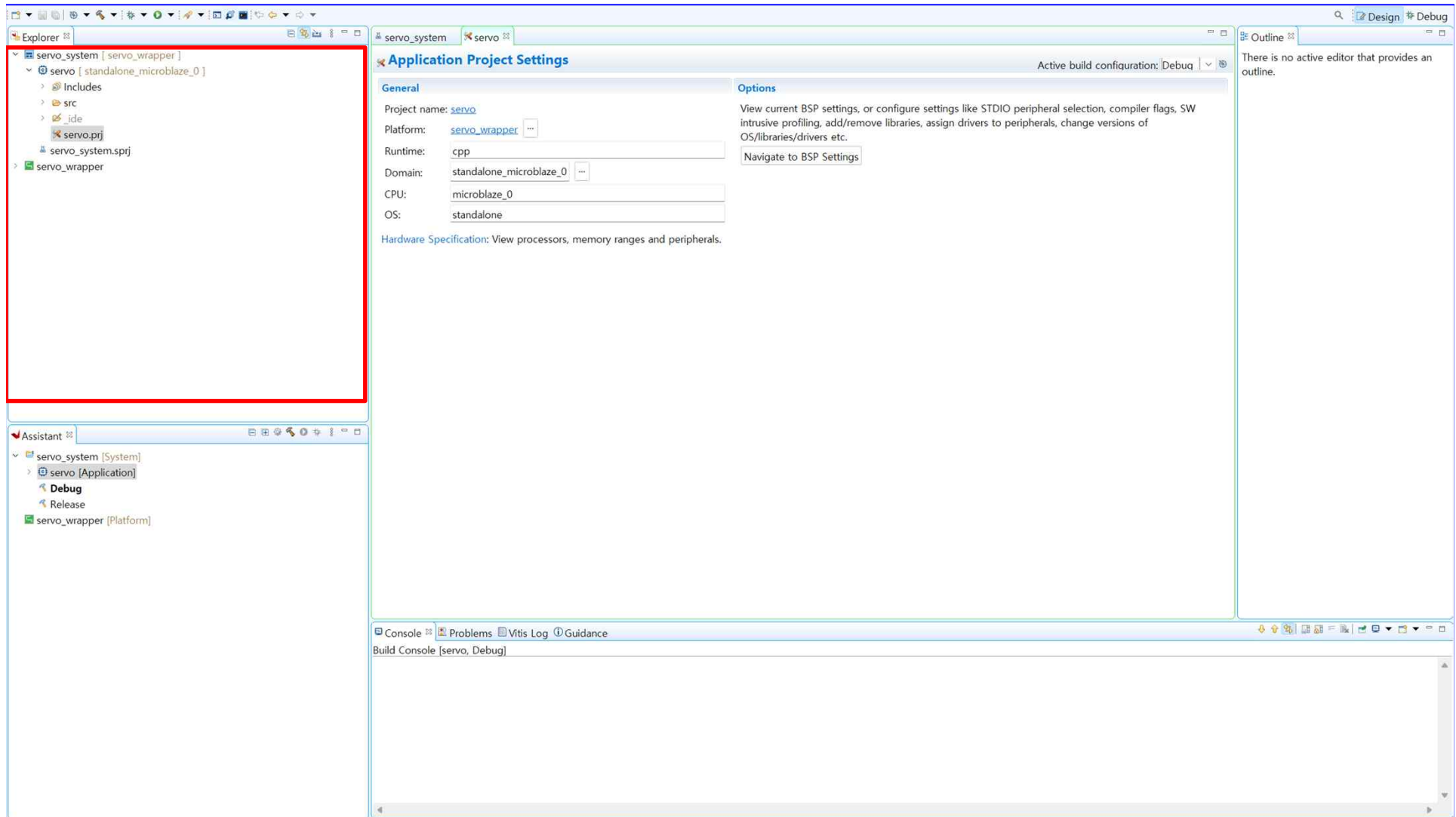
Show all processors in the hardware specification ☒

Next

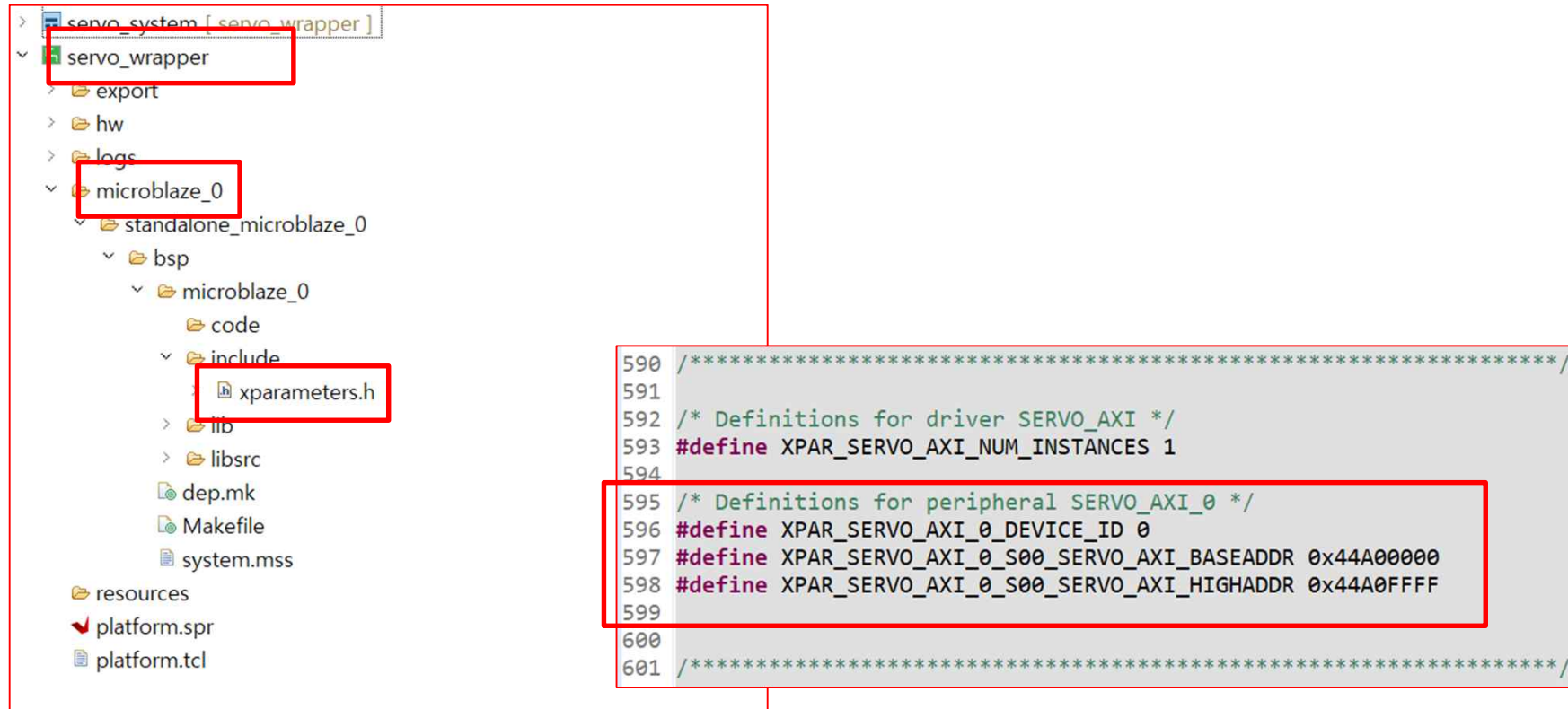
➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



➤ AXI Connect ➔ AXI Template



```
590 /*****  
591  
592 /* Definitions for driver SERVO_AXI */  
593 #define XPAR_SERVO_AXI_NUM_INSTANCES 1  
594  
595 /* Definitions for peripheral SERVO_AXI_0 */  
596 #define XPAR_SERVO_AXI_0_DEVICE_ID 0  
597 #define XPAR_SERVO_AXI_0_S00_SERVO_AXI_BASEADDR 0x44A00000  
598 #define XPAR_SERVO_AXI_0_S00_SERVO_AXI_HIGHADDR 0x44A0FFFF  
599  
600  
601 /*****/
```

BASEADDR 확인

➤ AXI Connect ➔ AXI Template

The screenshot displays the Xilinx IDE interface. On the left, the Explorer pane shows the project structure with 'helloworld.c' highlighted under the 'src' directory. The main editor window shows the source code for 'helloworld.c', which includes a copyright notice, a description of the application, and the main function. The code is as follows:

```
20 * LIMITED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,  
21 * FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL  
22 * XILINX BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY,  
23 * WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF  
24 * OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE  
25 * SOFTWARE.  
26 *  
27 * Except as contained in this notice, the name of the Xilinx shall not be used  
28 * in advertising or otherwise to promote the sale, use or other dealings in  
29 * this Software without prior written authorization from Xilinx.  
30 *  
31 *  
32 *  
33 /*  
34 * helloworld.c: simple test application  
35 *  
36 * This application configures UART 16550 to baud rate 9600.  
37 * PS7 UART (Zynq) is not initialized by this application, since  
38 * bootrom/bsp configures it to baud rate 115200  
39 *  
40 * -----  
41 * | UART TYPE   BAUD RATE |  
42 * -----  
43 * uarts550     9600  
44 * uartlite     Configurable only in HW design  
45 * ps7_uart     115200 (configured by bootrom/bsp)  
46 */  
47  
48 #include <stdio.h>  
49 #include "platform.h"  
50 #include "xil_printf.h"  
51  
52  
53 int main()  
54 {  
55     init_platform();  
56  
57     print("Hello World\n\r");  
58     print("Successfully ran Hello World application");  
59     cleanup_platform();  
60     return 0;  
61 }  
62
```

On the right, the Outline pane shows the file structure: 'stdio.h', 'platform.h', 'xil_printf.h', and 'main() : int'. The bottom status bar indicates 'Writable', 'Smart Insert', and '1 : 1 : 0'.

Vitis Coding

➤ AXI Connect ➔ AXI Template

Helloworld.c

```
#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xparameters.h"
#include "xgpio.h"
#include "sleep.h"
```

주소 확인

```
#define AXI_SERVO_BASEADDR 0x44A00000
```

```
/* definitions for btn0, AXI_GPIO_1 */
#define GPIO_BTN0_DEVICE_ID XPAR_GPIO_0_DEVICE_ID
#define BTN0_CHANNEL1 1
#define BTN0_INT_MSK XGPIO_IR_CH1_MASK

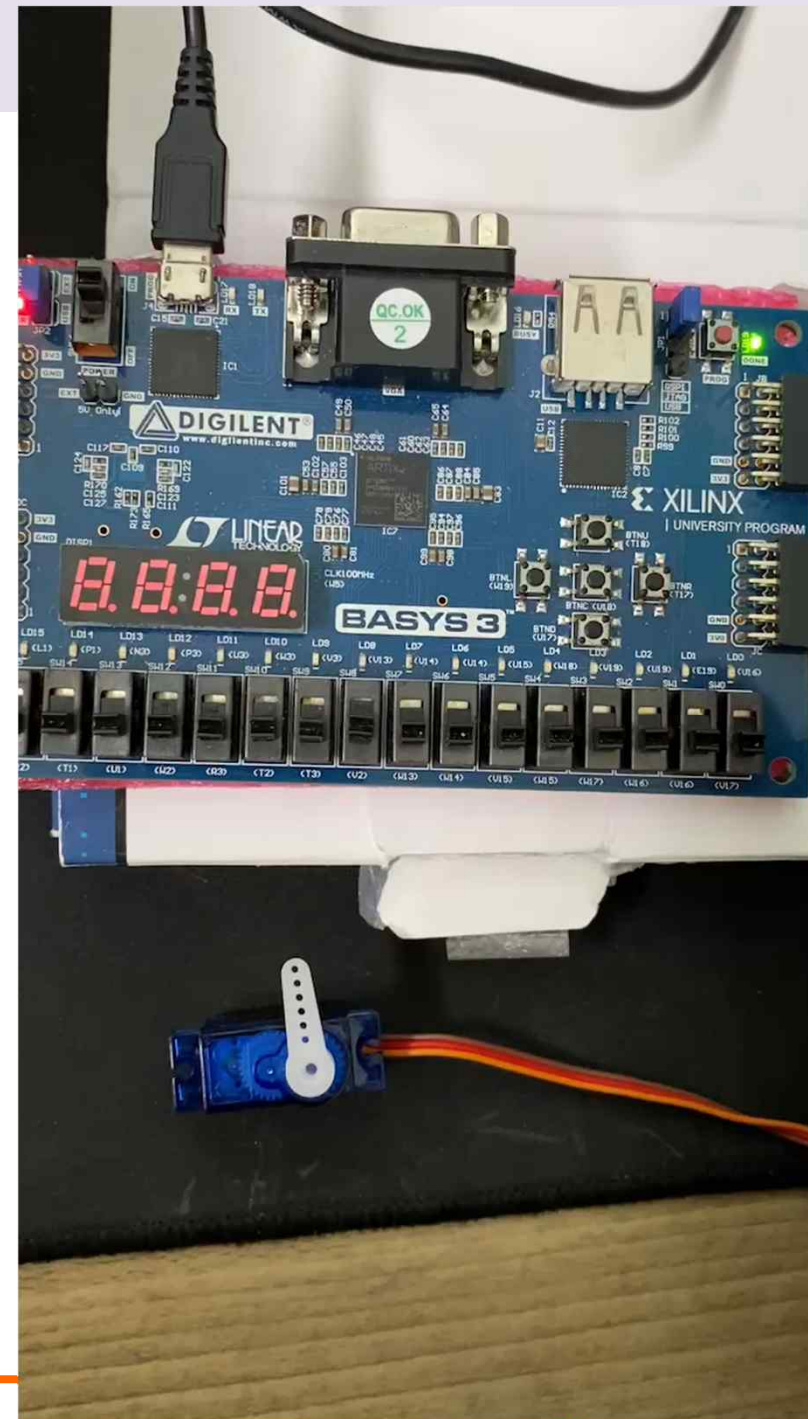
XGpio GPIO_BTN; /* The instance of the btn */
int main()
{
    init_platform();
    XGpio_Initialize(&GPIO_BTN, GPIO_BTN0_DEVICE_ID);
    print("Hello World\n\r");
    cleanup_platform();
}
```

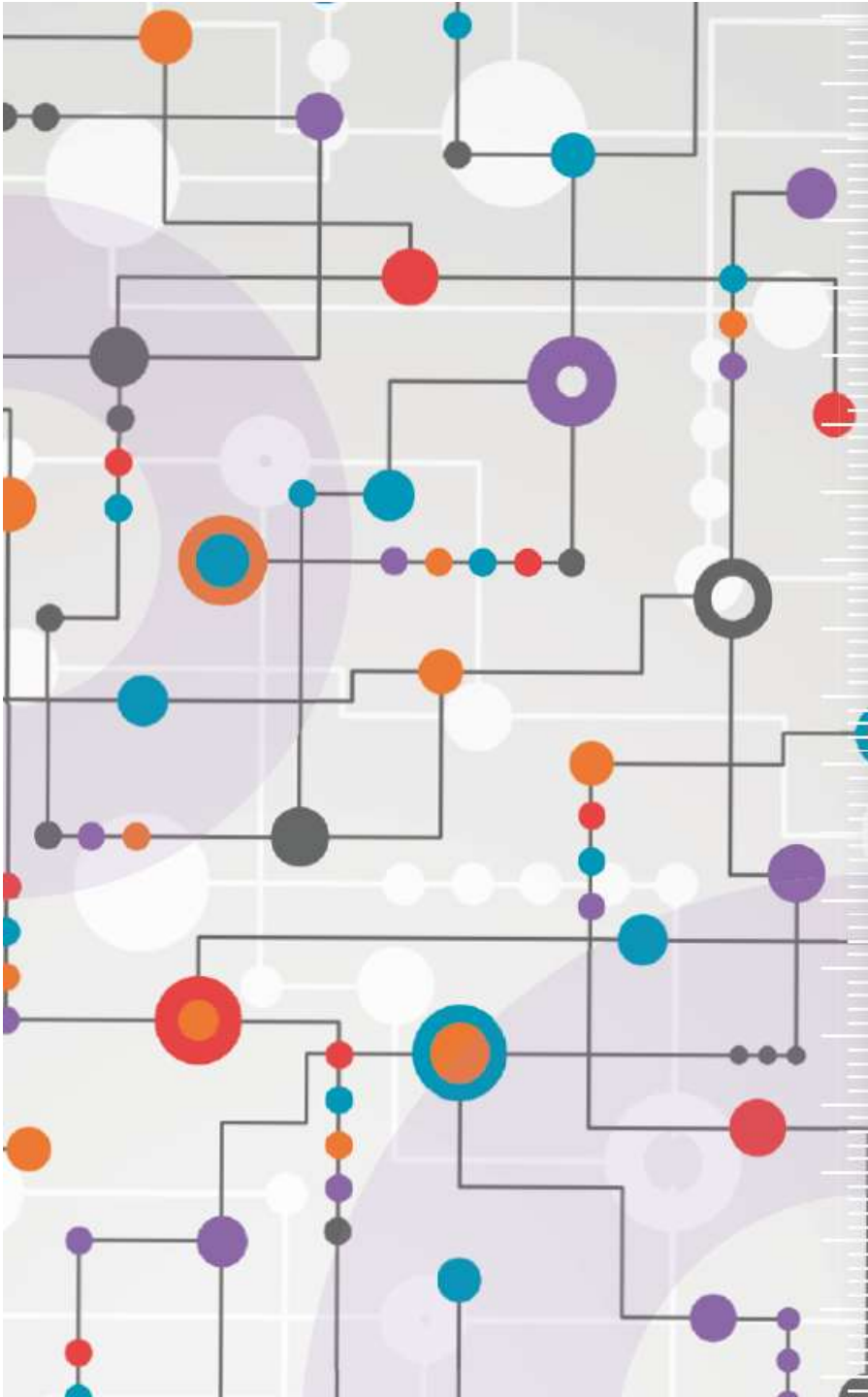
Helloworld.c

```
while(1){
    if ((XGpio_DiscreteRead(&GPIO_BTN,
        BTN0_CHANNEL1) & (1<<0)))
        Xil_Out32(AXI_SERVO_BASEADDR,0);
    else if ((XGpio_DiscreteRead(&GPIO_BTN,
        BTN0_CHANNEL1) & (1<<1)))
        Xil_Out32(AXI_SERVO_BASEADDR,7);
    else if ((XGpio_DiscreteRead(&GPIO_BTN,
        BTN0_CHANNEL1) & (1<<2)))
        Xil_Out32(AXI_SERVO_BASEADDR,16);
    else if ((XGpio_DiscreteRead(&GPIO_BTN,
        BTN0_CHANNEL1) & (1<<3)))
        Xil_Out32(AXI_SERVO_BASEADDR,26);

}
return 0;
}
```

➤ Check The Result





수고하셨습니다.